

Proofs of Quantum Memory

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Abstract

With the rapid advances in quantum computer architectures and the emerging prospect of large-scale quantum memory, it is becoming essential to classically verify that remote devices genuinely allocate the promised quantum memory with specified number of qubits and coherence time. In this paper, we introduce a new concept, *proofs of quantum memory (PoQM)*. A PoQM is an interactive protocol between a classical probabilistic polynomial-time (PPT) verifier and a quantum polynomial-time (QPT) prover over a classical channel where the verifier can verify that the prover has possessed a quantum memory with a certain number of qubits during a specified period of time. PoQM generalize the notion of proofs of quantumness (PoQ) [Brakerski, Christiano, Mahadev, Vazirani, and Vidick, JACM 2021]. Our main contributions are a formal definition of PoQM and its constructions based on hardness of LWE. Specifically, we give two constructions of PoQM. The first is of a four-round and has negligible soundness error under subexponential-hardness of LWE. The second is of a polynomial-round and has inverse-polynomial soundness error under polynomial-hardness of LWE. As a lowerbound of PoQM, we also show that PoQM imply one-way puzzles. Moreover, a certain restricted version of PoQM implies quantum computation classical communication (QCCC) key exchange.

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1 Introduction

Imagine a quantum computing startup claiming that it has built a quantum processor equipped with a 100-million-qubit quantum memory with a coherence time of 100 hours. How could a classical investor verify such a claim?

Proofs of quantumness (PoQ) [BCM⁺21] are insufficient for this purpose. A proof of quantumness is an interactive protocol between a classical probabilistic polynomial-time (PPT) verifier and a quantum polynomial-time (QPT) prover over a classical channel. Completeness is that if the prover behaves honestly, the verifier accepts with high probability, and soundness is that the verifier rejects with high probability if the prover is PPT. Using PoQ, a classical investor could confirm that the quantum startup is doing something at least non-classical, but it cannot verify that the startup can manipulate a 100-million-qubit quantum memory. Moreover, the classical investor cannot confirm that the quantum startup can keep the quantum coherence for 100 hours.

With the rapid advances in quantum computer architectures and the emerging prospect of large-scale quantum memory, it is becoming essential to classically verify that remote devices genuinely allocate the promised quantum memory with a specified number of qubits and coherence time. This motivates the following questions.

1. **Verification of the number of qubits:** *Can a classical verifier verify that a remote prover has possessed a quantum memory with a specified number of qubits?*
2. **Verification of the coherence time:** *Can a classical verifier verify that a remote prover has kept a quantum coherence for a specified period of time?*

1.1 Our Results

In this paper, we address both of these questions simultaneously by introducing a new concept, which we call *proofs of quantum memory* (PoQM). A PoQM is an interactive protocol between a PPT verifier and a QPT prover over a classical channel where the verifier can verify that the prover has kept a specified number of qubits during a specified time period.

Our contributions are summarized as follows:

1. We give a formal definition of PoQM.
2. We construct PoQM from the hardness of LWE.
3. We show lowerbounds of PoQM: PoQM imply one-way puzzles (OWPuzzs), which is a natural quantum analogue of one-way functions (OWFs) [KT24]. Moreover, a certain restricted version of PoQM implies quantum key-exchange over a classical channel.

In the following, we provide more details.

Formal definition of PoQM. We formally define PoQM as follows.¹(See Figure 1.) Let $\alpha, \beta : \mathbb{N} \rightarrow [0, 1]$ be any functions. Let $m_1, m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any (polynomially bounded²) functions. An $(\alpha, \beta, m_1, m_2)$ -PoQM $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ is a set of interactive algorithms over a classical channel. $\mathcal{V}_1$ and $\mathcal{V}_2$ are PPT, and $\mathcal{P}_1$ and $\mathcal{P}_2$ are QPT. The interaction consists of two phases, the initialization phase and the execution phase. In the

¹Our definition is based on (classical) proofs of space [DFKP15] and quantum proofs of space [MV20].

²This means $m_1, m_2 = O(\lambda^c)$ for some constant $c > 0$. This condition is occasionally omitted if it is clear from the context.

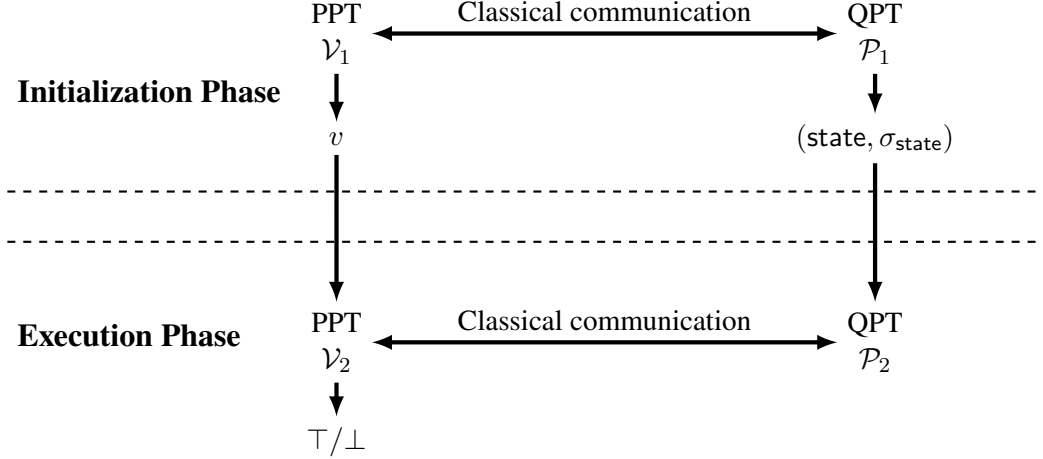


Figure 1: A PoQM consists of two phases: the initialization phase and the execution phase. At the end of the initialization phase, $\mathcal{V}_1$ outputs a classical bit string v , and $\mathcal{P}_1$ outputs a classical bit string state and an m_1 -qubit quantum state σ_{state} . At the beginning of the execution phase, $\mathcal{V}_2$ takes v as input, and $\mathcal{P}_2$ takes $(\text{state}, \sigma_{\text{state}})$ as input. At the end of the execution phase, $\mathcal{V}_2$ outputs $\top$ or $\perp$.

initialization phase, both $\mathcal{V}_1$ and $\mathcal{P}_1$ take the security parameter 1^λ as input, and interact over a classical channel. $\mathcal{V}_1$ outputs a classical bit string v , and $\mathcal{P}_1$ outputs a classical bit string state and an m_1 -qubit quantum state σ_{state} .³ In the execution phase, $\mathcal{V}_2$ takes v as input, and $\mathcal{P}_2$ takes $(\text{state}, \sigma_{\text{state}})$ as input. They interact over a classical channel, and $\mathcal{V}_2$ outputs $\top$ or $\perp$. α -completeness requires that $\mathcal{V}_2$ outputs $\top$ with probability at least α , that is, the honest prover with m_1 -qubit memory is accepted with high probability. On the other hand, (β, m_2) -soundness is defined as follows. Let $\mathcal{P}_1^*$ be a QPT algorithm that interacts with $\mathcal{V}_1$, and outputs a classical bit string s and an m_2 -qubit state ρ . Let $\mathcal{P}_2^*$ be a QPT algorithm that takes (s, ρ) as input, and interacts with $\mathcal{V}_2$. Then for any such $(\mathcal{P}_1^*, \mathcal{P}_2^*)$, $\mathcal{V}_2$ outputs $\top$ with probability at most β . This intuitively means that any malicious prover that can preserve at most m_2 -qubit quantum memory during the interval between the initialization phase and the execution phase cannot be accepted by the verifier. In other words, if the verifier accepts, the verifier can verify that the prover has possessed at least $(m_2 + 1)$ -qubit quantum memory during the interval between the initialization phase and the execution phase. Note that we do not make any upperbound for the size of the classical bit string s .

Relation to PoQ. We observe that PoQM generalize the notion of PoQ:

Theorem 1.1. *Let $\alpha, \beta : \mathbb{N} \rightarrow [0, 1]$ be any functions. Let $m_1 : \mathbb{N} \rightarrow \mathbb{N}$ be any function. If $(\alpha, \beta, m_1, 0)$ -PoQM exist, then (α, β) -PoQ exist.*

Here, an (α, β) -PoQ is a PoQ with completeness α and soundness β . Because an $(\alpha, \beta, m_1, m_2)$ -PoQM is trivially an $(\alpha, \beta, m_1, m_2 - 1)$ -PoQM for any $m_2 \geq 1$, we also obtain the following corollary.

Corollary 1.2. *Let $\alpha, \beta : \mathbb{N} \rightarrow [0, 1]$ be any functions. Let $m_1, m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any functions. If $(\alpha, \beta, m_1, m_2)$ -PoQM exist, then (α, β) -PoQ exist.*

³During the operation of the initialization phase, the prover may need more than m_1 qubits.

Constructions of PoQM. We give two constructions of PoQM based on the hardness of LWE.

The first construction is based on the subexponential hardness of LWE.

Theorem 1.3. *Let $m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any polynomially bounded function. Assuming the subexponential hardness of LWE, four-round $(1 - \text{negl}, \text{negl}, m_1, m_2)$ -PoQM exist with some polynomial m_1 .*

The second construction is based on the polynomial hardness of LWE.

Theorem 1.4. *Let p be any polynomial. Let $m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any polynomially bounded function such that $m_2(\lambda) = \omega(\log(\lambda))$. Assuming the polynomial hardness of LWE, r -round $(1 - \text{negl}, 1/p, \lceil 9.1m_2 \rceil, m_2)$ -PoQM exist with a certain polynomial r .*

These two results are incomparable. The first construction is of four-round and with negligible soundness, while the assumption, subexponential hardness of LWE, is stronger. On the other hand, the second construction is based on polynomial hardness of LWE, but it is of poly-round and soundness is only $1/\text{poly}$.⁴

Lowerbounds of PoQM. We show that one-way puzzles (OWPuzzs) [KT24] are a lowerbound of PoQM:

Theorem 1.5. *Let $\alpha, \beta : \mathbb{N} \rightarrow [0, 1]$ be any functions such that $\alpha(\lambda) - \beta(\lambda) \geq 1/\text{poly}(\lambda)$ for all sufficiently large $\lambda \in \mathbb{N}$. Let $m_1, m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any functions. If $(\alpha, \beta, m_1, m_2)$ -PoQM exist, then OWPuzzs exist.*

OWPuzzs are a natural quantum analogue of OWFs. A OWPuzz is a pair $(\text{Samp}, \text{Ver})$ of algorithms. Samp is a QPT algorithm that takes the security parameter 1^λ as input and outputs classical bit strings puzz and ans. Ver is an unbounded algorithm that takes $(\text{puzz}, \text{ans})$ as input and outputs $\top$ or $\perp$. The correctness is $\Pr[\top \leftarrow \text{Ver}(\text{puzz}, \text{ans}) : (\text{puzz}, \text{ans}) \leftarrow \text{Samp}(1^\lambda)] \geq 1 - \text{negl}(\lambda)$ and the security is $\Pr[\top \leftarrow \text{Ver}(\text{puzz}, \text{ans}') : (\text{puzz}, \text{ans}) \leftarrow \text{Samp}(1^\lambda), \text{ans}' \leftarrow \mathcal{A}(1^\lambda, \text{puzz})] \leq \text{negl}(\lambda)$ for any QPT adversary $\mathcal{A}$. OWPuzzs are implied by many quantum cryptographic primitives, and imply non-interactive bit commitment and multiparty computations [KT24, MY22, Yan22, AQY22].

If we consider a restricted version of PoQM (which we call *an extractable PoQM*), then we obtain a potentially stronger lowerbound:

Theorem 1.6. *Let $m_1, m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any functions. Let $\alpha : \mathbb{N} \rightarrow [0, 1]$ be any function. Let c_1 and c_2 be any constants such that $c_1 > c_2 > 0$. Let $p(\lambda) := \lambda^{c_1}$ and $q(\lambda) := \lambda^{c_2}$. If $(\alpha, \alpha - \frac{1}{q}, m_1, m_2)$ -extractable PoQM with extraction probability $1 - \frac{1}{p}$ exist, then quantum computation classical communication key-exchange (QCCC KE) exist.*

Here, an $(\alpha, \beta, m_1, m_2)$ -extractable PoQM with extraction probability γ is an $(\alpha, \beta, m_1, m_2)$ -PoQM where the execution phase is of a single round (i.e., of two message), and $\mathcal{P}_2$'s message in the execution phase can be computed in QPT by $\mathcal{V}_2$ with probability at least γ . Our construction of PoQM based on polynomial hardness of LWE satisfies this property. A QCCC KE is a key exchange in the quantum computation and classical communication (QCCC) setting, i.e., Alice and Bob are QPT but all communications are classical.

These two lowerbounds give interesting insights to the following two open problems about PoQ:

1. Is there any quantum cryptographic lowerbound for PoQ?⁵

⁴Parallel repetitions are non-trivial in PoQM, and we do not know how to do it.

⁵[MSY25] showed that PoQ imply classically-secure OWPuzzs, but we do not know any quantumly-secure cryptographic primitive implied by PoQ.

2. Can PoQ be constructed from OWFs?

Although we do not solve the first open problem in this paper, Theorem 1.5 at least shows that if we consider the generalization of PoQ (namely, PoQM), a meaningful lowerbound (namely, OWPuzzs) can be obtained. Moreover, Theorem 1.6 indicates that at least (a restricted version of) the generalization of PoQ (namely, extractable PoQM) will not be constructed from OWFs in a black-box way, because there is evidence that QCCC KE will not be constructed from OWFs in a black-box way [LLLL25, ACC⁺22, LLLL24].

1.2 Technical Overview

Here we provide a high-level overview of our results.

PoQM based on polynomial hardness of LWE. Let us first explain our construction of $(1 - \text{negl}, 1/p, \lceil 9.1m_2 \rceil, m_2)$ -PoQM for any polynomial p and any polynomially bounded function $m_2 : \mathbb{N} \rightarrow \mathbb{N}$ such that $m_2(\lambda) = \omega(\log(\lambda))$ based on polynomial hardness of LWE. The basic idea of our construction is simple: First, let us consider the following “information-theoretically-secure” $(1 - \text{negl}, \text{negl}, n, 0)$ -PoQM [MV20]:

- **Initialization phase.** $\mathcal{V}_1$ generates the state $\sigma := \bigotimes_{i=1}^n H^{\theta_i} |x_i\rangle$ with random $(x, \theta) \in \{0, 1\}^n \times \{0, 1\}^n$ by itself and sends the state to $\mathcal{P}_1$ over a quantum channel. (Here, x_i and θ_i are the i th bit of x and θ , respectively. H is the Hadamard operator.) $\mathcal{V}_1$ outputs (x, θ) . $\mathcal{P}_1$ outputs σ .
- **Execution phase.** $\mathcal{V}_2$ takes (x, θ) as input. $\mathcal{P}_2$ takes σ as input. $\mathcal{V}_2$ sends θ to $\mathcal{P}_2$, and $\mathcal{P}_2$ measures i th qubit of σ in the computational (Hadamard) basis if $\theta_i = 0$ ($\theta_i = 1$) for all $i \in [n]$. Let x'_i be the measurement result on the i th qubit. If $x_i = x'_i$ for all $i \in [n]$, $\mathcal{V}_2$ accepts. Otherwise, $\mathcal{V}_2$ rejects.

This information-theoretically-secure PoQM does not achieve our goal, because of the following two reasons:

1. It is only the case when $m_2 = 0$. We want to construct $(1 - \text{negl}, 1/\text{poly}, \lceil 9.1m_2 \rceil, m_2)$ -PoQM for any polynomial m_2 .
2. Both the verifier and the channel are quantum.

The first issue is solved by using a lemma of [BZ13]. The lemma says the following: Let $\mathcal{A}$ be a quantum algorithm that outputs a classical bit string. Let $\mathcal{A}'$ be the algorithm that is the same as $\mathcal{A}$ except that a k -outcome measurement is done at any step. Then, $\Pr[x \leftarrow \mathcal{A}] \leq k \Pr[x \leftarrow \mathcal{A}']$ for any x .⁶ Using this lemma, we can show that an $(\alpha, \beta, m_1, 0)$ -PoQM is an $(\alpha, 2^{m_2}\beta, m_1, m_2)$ -PoQM for any α, β, m_1, m_2 : Let $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ be an $(\alpha, \beta, m_1, 0)$ -PoQM. Assume that it is not an $(\alpha, 2^{m_2}\beta, m_1, m_2)$ -PoQM. Then there exists a pair $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ of QPT adversaries such that $\mathcal{P}_1^*$ outputs a classical bit string s and an m_2 -qubit state ρ , $\mathcal{P}_2^*$ takes (s, ρ) as input, and the probability that $\mathcal{V}_2$ outputs $\top$ is strictly larger than $2^{m_2}\beta$. Let us define another QPT adversary $\mathcal{P}_1^{**}$ as follows: $\mathcal{P}_1^{**}$ runs $\mathcal{P}_1^*$ but it measures all qubits of the output state ρ of $\mathcal{P}_1^*$ in the computational basis, and outputs the measurement result. Then, by using the lemma of [BZ13], the probability that $(\mathcal{P}_1^{**}, \mathcal{P}_2^*)$ is accepted is strictly larger than $2^{-m_2} \times 2^{m_2}\beta = \beta$, which is the contradiction.

Therefore, we want to show that soundness (i.e., β) of the above information-theoretically-secure PoQM is $\beta = 2^{-m_2} \times \text{negl}$.⁷ In order to show it, we use the lemma, *the LOCC leakage property for BB84 states*, which was introduced in [CG24] for another purpose, namely, constructions of leakage-resilient encryption and signatures. This lemma shows the following. Let us consider the following security game:

⁶ $\Pr[x \leftarrow \mathcal{A}]$ is the probability that $\mathcal{A}$ outputs x .

⁷At this stage, we can achieve negl -soundness, but due to the $1/\text{poly}$ -soundness of RSPs, what we finally get is only $1/\text{poly}$ -soundness.

1. A challenger generates a random BB84 state $\sigma := \bigotimes_{i=1}^n H^{\theta_i} |x_i\rangle$ with random $(x, \theta) \in \{0, 1\}^n \times \{0, 1\}^n$.
2. An adversary sends the challenger a classical description E of a quantum algorithm that takes a quantum state as input and outputs a classical bit string.
3. The challenger runs $\eta \leftarrow E(\sigma)$ ⁸, and sends the classical bit string η to the adversary.
4. The challenger sends θ to the adversary.
5. The adversary returns a bit string $x' \in \{0, 1\}^n$ to the challenger.
6. The challenger accepts if $x = x'$, and rejects if $x \neq x'$.

The lemma says that the probability that the challenger accepts is at most $2^{-\frac{\xi}{2} \cdot n + 2^{-n}}$, where $\xi := -\log\left(\frac{1}{2} + \frac{1}{2\sqrt{2}}\right) > 0.22$. By considering the above η as the output of $\mathcal{P}_1^*$, and taking $n = \lceil 9.1m_2 \rceil$, we can show that the probability that the verifier accepts in the information-theoretically-secure PoQM is $2^{-\frac{\xi}{2} \lceil 9.1m_2 \rceil + 2^{-\lceil 9.1m_2 \rceil}} \leq 2^{-m_2} \times 2^{-0.001m_2 + 2^{-9.1m_2}} = 2^{-m_2} \times \text{negl}$. (Note that $m_2(\lambda) = \omega(\log(\lambda))$ by assumption.)

The second issue that both the verifier and the channel are quantum in the information-theoretically-secure PoQM is solved by using verifiable remote state preparations (RSPs) [GV19, Zha25]. An RSP is a two party protocol between a PPT sender and a QPT receiver. They interact over a classical channel. The receiver outputs a random BB84 state, and the sender outputs its classical description. By using this, we can replace the quantum verifier and the quantum channel with a PPT verifier and a classical channel. Because the RSP of [Zha25] requires poly round of communication, and it achieves only 1/poly-soundness, the final PoQM we obtain is of poly round, and has only 1/poly-soundness.

PoQM from subexponential hardness of LWE. The above construction requires polynomial rounds of communication and achieves only 1/poly-soundness. To complement this, for any polynomially bounded function $m_2 : \mathbb{N} \rightarrow \mathbb{N}$, we next construct four-round $(1 - \text{negl}, \text{negl}, m_1, m_2)$ -PoQM with some polynomial m_1 from subexponential hardness of LWE.

The construction is based on 1-of- 2^k puzzles [LLQ22]. 1-of-2 puzzles were first introduced in [RS19] to study *semi-quantum money*, which is a variant of quantum money that can be minted and verified classically. [LLQ22] extended 1-of-2 puzzles to 1-of- 2^k puzzles to construct classically verifiable position verification. A 1-of- 2^k puzzle consists of four algorithms (KeyGen, Obligate, Solve, Ver). KeyGen is a PPT algorithm that takes the security parameter 1^λ as input and outputs a public key pk and a secret key sk . Obligate is a QPT algorithm that takes pk as input and outputs a bit string y and a quantum state ρ . Solve is a QPT algorithm that takes pk , y , ρ and a randomly chosen k -bit string ch as input and outputs a classical answer ans . Ver is a polynomial-time classical deterministic algorithm that takes sk , y , ch and ans as input and outputs $\top$ or $\perp$. Completeness is that Ver outputs $\top$ with probability at least $1 - \text{negl}(\lambda)$. In order to define soundness, we consider the following security game between a set $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ of adversaries and a challenger Chal.

1. Chal runs $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$.
2. $\mathcal{A}$ receives the public key pk and outputs a bit string y and a quantum state $\sigma_{\mathbf{BC}}$ over two registers $\mathbf{B}$ and $\mathbf{C}$.
3. $\mathcal{A}$ sends y to Chal. $\mathcal{A}$ sends the register $\mathbf{B}$ to $\mathcal{B}$. $\mathcal{A}$ sends the register $\mathbf{C}$ to $\mathcal{C}$.

⁸ $\eta \leftarrow E(\sigma)$ means that the algorithm E is run on input σ , and the output is η .

4. Chal samples $\text{ch} \leftarrow \{0, 1\}^{k(\lambda)}$ and sends ch to both $\mathcal{B}$ and $\mathcal{C}$.
5. $\mathcal{B}$ outputs an answer $\text{ans}_{\mathcal{B}}$, and sends it to Chal. $\mathcal{C}$ outputs an answer $\text{ans}_{\mathcal{C}}$, and sends it to Chal.
6. Chal outputs $\top$ if and only if

$$\text{Ver}(\text{sk}, y, \text{ch}, \text{ans}_{\mathcal{B}}) = \top \wedge \text{Ver}(\text{sk}, y, \text{ch}, \text{ans}_{\mathcal{C}}) = \top. \quad (1)$$

With this security game, we define c -soundness as follows: for any set $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ of non-uniform QPT adversaries,

$$\Pr[\top \leftarrow \text{Chal}] \leq 2^{-k(\lambda)} + \text{negl}(2^{\lambda^c}). \quad (2)$$

We want to construct a four-round $(1 - \text{negl}, \text{negl}, m_1, m_2)$ -PoQM. Let $c > 0$ be any constant such that $m_2(\lambda) = O(\lambda^c)$. Set $k(\lambda) = \omega(\lambda^c)$. We construct a four-round $(1 - \text{negl}, \text{negl}, m_1, m_2)$ -PoQM from 1-of- 2^k puzzles with c -soundness as follows.

- **Initialization phase.** $\mathcal{V}_1$ runs $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$ and sends pk to $\mathcal{P}_1$. $\mathcal{P}_1$ runs $(y, \rho) \leftarrow \text{Obligate}(\text{pk})$ and sends y to $\mathcal{V}_1$. $\mathcal{V}_1$ outputs (sk, y) , and $\mathcal{P}_1$ outputs (pk, y, ρ) .
- **Execution phase.** $\mathcal{V}_2$ takes (sk, y) as input, and $\mathcal{P}_2$ takes (pk, y, ρ) as input. $\mathcal{V}_2$ samples random k -bit string ch and sends it to $\mathcal{P}_2$. $\mathcal{P}_2$ runs $\text{ans} \leftarrow \text{Solve}(\text{pk}, y, \rho, \text{ch})$, and sends ans to $\mathcal{V}_2$. $\mathcal{V}_2$ runs $\top/\perp \leftarrow \text{Ver}(\text{sk}, y, \text{ch}, \text{ans})$ and outputs the output.

Thus constructed PoQM is $(1 - \text{negl}, \epsilon, m_1, 0)$ -PoQM with $\epsilon(\lambda) = (2^{-k(\lambda)} + \text{negl}(2^{\lambda^c}))^{\frac{1}{2}}$, where $m_1(\lambda)$ denotes the length of the output quantum state ρ of Obligate. The reason is as follows. Assume that it is not $(\epsilon, 0)$ -sound. Then, there exists a pair $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ of QPT adversaries such that $\mathcal{P}_1^*$ outputs only a classical bit string s , $\mathcal{P}_2^*$ takes only the classical bit string as input, and $\mathcal{V}_2$ outputs $\top$ with probability at least ϵ . From such $(\mathcal{P}_1^*, \mathcal{P}_2^*)$, we can construct a set $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ of adversaries for the 1-of- 2^k puzzle whose winning probability is strictly larger than ϵ^2 as follows: Given pk , $\mathcal{A}$ runs $s \leftarrow \mathcal{P}_1^*(\text{pk})$, sends s to both $\mathcal{B}$ and $\mathcal{C}$. $\mathcal{B}$ and $\mathcal{C}$ run $\text{ans} \leftarrow \mathcal{P}_2^*(s, \text{ch})$ and send ans to Chal, respectively. Because $\epsilon^2(\lambda) = ((2^{-k(\lambda)} + \text{negl}(2^{\lambda^c}))^{\frac{1}{2}})^2 = 2^{-k(\lambda)} + \text{negl}(2^{\lambda^c})$, c -soundness is broken.

Assuming subexponential hardness of LWE, for any constant $c > 0$ and for any polynomial k , there exist 1-of- 2^k puzzles with c -soundness [LLQ22].

Finally, by using the lemma of [BZ13], thus constructed $(1 - \text{negl}, \epsilon, m_1, 0)$ -PoQM is $(1 - \text{negl}, 2^{m_2}\epsilon, m_1, m_2)$ -PoQM. Because $2^{m_2(\lambda)}\epsilon(\lambda) = 2^{O(\lambda^c)}\text{negl}(2^{\lambda^c}) = \text{negl}(2^{\lambda^c}) = \text{negl}(\lambda)$, we finally obtain $(1 - \text{negl}, \text{negl}, m_1, m_2)$ -PoQM.

PoQM imply OWPuzzs. As a lowerbound of PoQM, we show that PoQM imply OWPuzzs. Because OWPuzzs are existentially equivalent to state puzzles (StatePuzzs) [KT25], we actually construct StatePuzzs from the PoQM. A StatePuzz is a QPT algorithm Samp that takes 1^λ as input and outputs a pair $(s, |\psi_s\rangle)$ of a bit string s and a pure quantum state $|\psi_s\rangle$. The security is that given s , no QPT algorithm can output a quantum state that is close to $|\psi_s\rangle$. Our construction of StatePuzzs from the PoQM is as follows: The classical puzzle s of the StatePuzz is the classical output state of $\mathcal{P}_1$ and the transcript τ of the initialization phase. The quantum answer $|\psi_s\rangle$ of the StatePuzz is the quantum output σ_{state} of $\mathcal{P}_1$. Intuitively, if thus constructed StatePuzz is not secure, there exists a QPT adversary $\mathcal{A}$ that, given $s = (\text{state}, \tau)$, can output a quantum state ρ that is close to σ_{state} . Then, the following adversary $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ can break the soundness of the PoQM: $\mathcal{P}_1^*$ outputs $s = (\text{state}, \tau)$. $\mathcal{P}_2^*$ takes s as input, runs $\rho \leftarrow \mathcal{A}(s)$, and runs $\mathcal{P}_2$ on (state, ρ) .

Extractable PoQM imply QCCC KE. We also show that, extractable PoQM imply QCCC KE. Our construction of QCCC KE is as follows: Alice and Bob run the initialization phase of the PoQM: Alice runs $\mathcal{V}_1$ and Bob runs $\mathcal{P}_1$. Alice and Bob next run the execution phase of the PoQM: Alice runs $\mathcal{V}_2$ and Bob runs $\mathcal{P}_2$. However, Bob does not send the last message of $\mathcal{P}_2$ to Alice. Because of the extractable property, Alice can compute Bob’s last message with high probability. This last message is used as the shared key, which shows the correctness of the KE. To show the security, assume that there is a QPT adversary Eve who, given the transcript of the interaction between Alice and Bob, can compute Alice’s key. By using such Eve, we can construct an adversary $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ that breaks the soundness of the extractable PoQM as follows: $\mathcal{P}_1^*$ outputs $s = (\text{state}, \tau)$, where τ is the transcript of the initialization phase. $\mathcal{P}_2^*$ takes s as input, runs Eve on input s and $\mathcal{V}_2$ ’s message, and outputs the output of Eve.

1.3 Related Works

[MM25] constructed information-theoretically-sound PoQ that are sound against *classical*-memory-bounded classical provers. They also constructed information-theoretically-secure claw generation that are secure against quantum-memory-bounded quantum provers. [CH22] introduced classical verification of quantum depth. [BGKM⁺23] constructed a test of qubit protocol. [CR20] constructed information-theoretically-sound quantum dimension test. Although these results share similar motivations, they would not be able to classically verify that a prover has possessed a certain amount of quantum memory during a specified period of time.

[MV20] introduced quantum proofs of space, which are a quantum variant of proofs of space [DFKP15]. Our definition of PoQM is based on them. The verifier or the channel of [MV20] is, however, quantum in their definitions and constructions.

2 Preliminaries

2.1 Basic Notations

We use standard notations of quantum computing and cryptography. All polynomials in this paper have coefficients in $\mathbb{N}$. We use λ as the security parameter. For a bit string x , x_i denotes the i th bit of x . For two bit strings x and y , $x||y$ means the concatenation of them. $[n]$ means the set $\{1, 2, \dots, n\}$. $\lceil x \rceil$ means the minimum integer greater than or equal to x . negl is a negligible function, and poly is a polynomial. PPT stands for (classical) probabilistic polynomial-time and QPT stands for quantum polynomial-time. We refer to a non-uniform QPT algorithm as a QPT algorithm with polynomial-size quantum advice. For a set S , $x \leftarrow S$ means that an element x is chosen from S uniformly at random. For an algorithm $\mathcal{A}$, $y \leftarrow \mathcal{A}(x)$ means that the algorithm $\mathcal{A}$ outputs y on input x . For an algorithm $\mathcal{A}$ that takes a quantum state as input and outputs a quantum state, $\mathcal{A}(\rho)$ often means the output state of $\mathcal{A}$ on input ρ . For two density matrices ρ and σ , $\text{TD}(\rho, \sigma)$ is their trace distance. For two interactive algorithms $\mathcal{A}$ and $\mathcal{B}$ over a classical channel, $\rho_{\mathbf{A}, \mathbf{B}} \leftarrow \langle \mathcal{A}(x), \mathcal{B}(y) \rangle$ means that $\mathcal{A}$ and $\mathcal{B}$ are executed on input x and y , respectively, and the final output state is a quantum state $\rho_{\mathbf{A}, \mathbf{B}}$ over two registers $\mathbf{A}$ and $\mathbf{B}$, where $\mathcal{A}$ ’s output register is $\mathbf{A}$ and $\mathcal{B}$ ’s output register is $\mathbf{B}$.

2.2 Lemmata

Here we explain two lemmata that we will use later.

Lemma 2.1 ([BZ13], Lemma 1). *Let $\mathcal{A}$ be a quantum algorithm that outputs a classical bit string. Let $\mathcal{A}'$ be another quantum algorithm obtained from $\mathcal{A}$ by pausing $\mathcal{A}$ at an arbitrary stage of the execution, performing*

a measurement that obtains one of k outcomes, and then resuming $\mathcal{A}$. Then $\Pr[x \leftarrow \mathcal{A}'] \geq \Pr[x \leftarrow \mathcal{A}]/k$ for any bit string x .

Lemma 2.2 (LOCC Leakage Property for BB84 States [CG24], Theorem 10). *Let us consider the following game between a (not-necessarily-polynomial-time) adversary $\mathcal{A}$ and a challenger $\mathcal{C}$:*

1. $\mathcal{C}$ samples $x, \theta \leftarrow \{0, 1\}^\lambda$ and outputs $|R_0\rangle := \bigotimes_{i=1}^\lambda H^{\theta_i} |x_i\rangle$. Here, H is the Hadamard operator.
2. Let $N : \mathbb{N} \rightarrow \mathbb{N}$ be a function. For $i = 1, 2, \dots, N(\lambda)$, $\mathcal{A}$ and $\mathcal{C}$ do the following.
 - (a) $\mathcal{A}$ sends $\mathcal{C}$ a classical description E_i of a (not-necessarily-polynomial-time) quantum algorithm which takes a quantum state as input and outputs a classical bit string and a pure quantum state.
 - (b) $\mathcal{C}$ runs the algorithm E_i on input $|R_{i-1}\rangle$. Let $(L_i, |R_i\rangle)$ be the output, where L_i is a classical bit string.
 - (c) $\mathcal{C}$ sends L_i to $\mathcal{A}$.
3. $\mathcal{C}$ sends θ to $\mathcal{A}$.
4. $\mathcal{A}$ outputs x' and sends it to $\mathcal{C}$.
5. $\mathcal{C}$ outputs $\top$ if $x' = x$, and otherwise it outputs $\perp$.

Then, for all sufficiently large $\lambda \in \mathbb{N}$ and for any (not-necessarily-polynomial-time) adversary $\mathcal{A}$,

$$\Pr[\top \leftarrow \mathcal{C}] \leq 2^{-\frac{\xi}{2} \cdot \lambda + 2^{-\lambda}}, \quad (3)$$

where $\xi := -\log\left(\frac{1}{2} + \frac{1}{2\sqrt{2}}\right) > 0.22$.

2.3 Cryptography

In this subsection, we explain several cryptographic primitives that we will use.

First, we recall the definition of proofs of quantumness (PoQ) introduced by [BCM⁺21].

Definition 2.3 (Proofs of Quantumness (PoQ) [BCM⁺21]). *An (α, β) -proof of quantumness (PoQ) is a set $(\mathcal{V}, \mathcal{P})$ of interactive algorithms over a classical channel. $\mathcal{V}$ (verifier) is a PPT algorithm that takes 1^λ as input and outputs $\top$ or $\perp$. $\mathcal{P}$ (prover) is a QPT algorithm that takes 1^λ as input and outputs nothing. We require the following two properties.*

α -completeness: *For all sufficiently large $\lambda \in \mathbb{N}$,*

$$\Pr[\top \leftarrow \langle \mathcal{V}(1^\lambda), \mathcal{P}(1^\lambda) \rangle] \geq \alpha(\lambda). \quad (4)$$

β -soundness: *For any non-uniform PPT adversary $\mathcal{P}^*$ and for all sufficiently large $\lambda \in \mathbb{N}$,*

$$\Pr[\top \leftarrow \langle \mathcal{V}(1^\lambda), \mathcal{P}^*(1^\lambda) \rangle] \leq \beta(\lambda). \quad (5)$$

Next, we give the definition of state puzzles (StatePuzzs).

Definition 2.4 (State Puzzles [KT25]). *Let $\epsilon : \mathbb{N} \rightarrow [0, 1]$ be a function. An ϵ -StatePuzz is a QPT algorithm Samp that takes 1^λ as input and outputs a pair $(s, |\psi_s\rangle)$ of a bit string s and a pure quantum state $|\psi_s\rangle$ satisfying the following property.*

Security: For any non-uniform QPT adversary $\mathcal{A}$ that takes s as input and outputs a quantum state, and for all sufficiently large $\lambda \in \mathbb{N}$,

$$\mathbb{E}_{(s, |\psi_s\rangle) \leftarrow \text{Samp}(1^\lambda)} \langle \psi_s | \mathcal{A}(s) | \psi_s \rangle \leq 1 - \epsilon(\lambda). \quad (6)$$

If $\epsilon(\lambda) = 1 - \text{negl}(\lambda)$, we just call it a state puzzle.

The following lemma is implicitly shown in [KT25].

Lemma 2.5. Let p be any polynomial. If $1/p$ -StatePuzzs exist, then StatePuzzs exist.

We also define 1-of- 2^k puzzles [LLQ22].

Definition 2.6 (1-of- 2^k puzzles [LLQ22]). Let k be a polynomial. A 1-of- 2^k puzzle is a set (KeyGen, Obligate, Solve, Ver) of algorithms with the following syntax.

- KeyGen(1^λ) $\rightarrow$ (pk, sk) : A PPT algorithm that takes 1^λ as input and outputs a public key pk and a secret key sk.
- Obligate(pk) $\rightarrow$ (y, ρ) : A QPT algorithm that takes pk as input and outputs a bit string y and a quantum state ρ .
- Solve(pk, y, ρ , ch) $\rightarrow$ ans : A QPT algorithm that takes pk, y, ρ and a challenge k -bit string ch as input and outputs a classical answer ans.
- Ver(sk, y , ch, ans) $\rightarrow$ $\top/\perp$: A polynomial-time classical deterministic algorithm that takes sk, y , ch and ans as input and outputs $\top$ or $\perp$.

We require the following properties.

Completeness:

$$\Pr \left[\top \leftarrow \text{Ver}(\text{sk}, y, \text{ch}, \text{ans}) : \begin{array}{l} (\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda) \\ (y, \rho) \leftarrow \text{Obligate}(\text{pk}) \\ \text{ch} \leftarrow \{0, 1\}^{k(\lambda)} \\ \text{ans} \leftarrow \text{Solve}(\text{pk}, y, \rho, \text{ch}) \end{array} \right] \geq 1 - \text{negl}(\lambda). \quad (7)$$

c-soundness: Let us consider the following game between a set $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ of adversaries and a challenger Chal:

1. Chal runs $(\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda)$.
2. $\mathcal{A}$ receives the public key pk, and outputs a bit string y and a quantum state $\sigma_{\mathbf{BC}}$ over two registers $\mathbf{B}$ and $\mathbf{C}$.
3. $\mathcal{A}$ sends y to Chal. $\mathcal{A}$ sends $\mathbf{B}$ the register $\mathbf{B}$. $\mathcal{A}$ sends $\mathbf{C}$ the register $\mathbf{C}$.
4. Chal samples $\text{ch} \leftarrow \{0, 1\}^{k(\lambda)}$ and sends ch to both $\mathcal{B}$ and $\mathcal{C}$.
5. $\mathcal{B}$ outputs an answer $\text{ans}_{\mathcal{B}}$ and sends it to Chal. $\mathcal{C}$ outputs an answer $\text{ans}_{\mathcal{C}}$ and sends it to Chal.

6. Chal outputs $\top$ if

$$\text{Ver}(\text{sk}, y, \text{ch}, \text{ans}_{\mathcal{B}}) = \top \wedge \text{Ver}(\text{sk}, y, \text{ch}, \text{ans}_{\mathcal{C}}) = \top. \quad (8)$$

Otherwise, Chal outputs $\perp$.

For any set $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ of non-uniform QPT adversaries,

$$\Pr[\top \leftarrow \text{Chal}] \leq 2^{-k(\lambda)} + \text{negl}(2^{\lambda^c}). \quad (9)$$

The following lemma is implicitly shown in [LLQ22].⁹

Lemma 2.7. *Assuming the subexponential hardness of LWE, for any $c > 0$ and for any polynomial k , 1-of- 2^k puzzles with c -soundness exist.*

We also use verifiable remote state preparations [GV19, Zha25]. In this paper, we use the formalism of [Zha25].

Definition 2.8 (Remote State Preparations (RSPs) [Zha25]). *Let $n : \mathbb{N} \rightarrow \mathbb{N}$ be any polynomially bounded function. Let p be polynomial. An $(n, \frac{1}{p})$ -remote state preparation (RSP) is a set $(\mathcal{V}, \mathcal{P})$ of interactive algorithms over a classical channel. $\mathcal{V}$ is a PPT algorithm that takes 1^λ as input and outputs classical bit strings $(x, \theta) \in \{0, 1\}^n \times \{0, 1\}^n$ and $\text{flag} \in \{\text{pass}, \text{fail}\}$. $\mathcal{P}$ is a QPT algorithm that takes 1^λ as input and outputs a quantum state on the register $\mathbf{Q}$. We require the following two properties.*

Completeness:

$$\text{TD}(\phi_{\mathbf{F}, \mathbf{D}, \mathbf{Q}}, |\text{pass}\rangle \langle \text{pass}|_{\mathbf{F}} \otimes \eta_{\mathbf{D}, \mathbf{Q}}) \leq \text{negl}(\lambda). \quad (10)$$

Here, for the notational simplicity, we consider that $\mathcal{V}$'s classical outputs are encoded in a quantum state in the computational basis. $\mathcal{V}$'s classical output flag is written in the register $\mathbf{F}$, and (x, θ) is written in the register $\mathbf{D}$. $\phi_{\mathbf{F}, \mathbf{D}, \mathbf{Q}} \leftarrow \langle \mathcal{V}(1^\lambda), \mathcal{P}(1^\lambda) \rangle$ and $\eta_{\mathbf{D}, \mathbf{Q}} := \frac{1}{4^n} \sum_{(x, \theta) \in \{0, 1\}^n \times \{0, 1\}^n} |x, \theta\rangle \langle x, \theta|_{\mathbf{D}} \otimes (\bigotimes_{i=1}^n H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i})_{\mathbf{Q}}$.

$\frac{1}{p}$ -soundness: *For any non-uniform QPT adversary $\mathcal{P}^*$ that outputs a quantum state on a register $\mathbf{Q}'$, there exists a non-uniform QPT algorithm Sim that maps a quantum state on the register $\mathbf{Q}$ to a quantum state on the registers $\mathbf{F}$ and $\mathbf{Q}'$ such that for any non-uniform QPT algorithm $\mathcal{D}$,*

$$\left| \text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'}] \Pr\left[\top \leftarrow \mathcal{D}\left(\frac{\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'} \Pi_{\mathbf{F}}^{\text{pass}}}{\text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'}]}\right)\right] \right. \quad (11)$$

$$\left. - \text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}})] \Pr\left[\top \leftarrow \mathcal{D}\left(\frac{\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}}) \Pi_{\mathbf{F}}^{\text{pass}}}{\text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}})]}\right)\right] \right| \leq \frac{1}{p(\lambda)}. \quad (12)$$

Here, $\Pi_{\mathbf{F}}^{\text{pass}} := |\text{pass}\rangle \langle \text{pass}|_{\mathbf{F}}$ and $\sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'} \leftarrow \langle \mathcal{V}(1^\lambda), \mathcal{P}^*(1^\lambda) \rangle$.

⁹[LLQ22] implicitly showed that, for any $c > 0$, assuming c -subexponential hardness of LWE, 1-of- 2^k puzzles with c -soundness exist. c -subexponential hardness of LWE roughly means that any quantum algorithm running in time $O(2^{\lambda^c})$ can distinguish two distributions with probability at most $\text{negl}(2^{\lambda^c})$. Let $c' > 0$ be any constant. By replacing the security parameter λ with $\lambda' := \lambda^{\frac{c}{c'}}$, c' -subexponential hardness of LWE can be converted to c -subexponential hardness of LWE. Thus, for any $c, c' > 0$, assuming c' -subexponential hardness of LWE, 1-of- 2^k puzzles with c -soundness exist.

The following lemma is shown in [Zha25]:

Lemma 2.9. *Assuming the polynomial hardness of LWE, for any polynomially bounded function $n : \mathbb{N} \rightarrow \mathbb{N}$ and polynomial p , r -round $(n, \frac{1}{p})$ -RSPs exist with a certain polynomial r .*

Finally, we explain quantum computation and classical communication key exchange (QCCC KE).

Definition 2.10 (QCCC Key Exchange (QCCC KE) [GMMY24]). *An (α, β) -QCCC key exchange (KE) is a set $(\mathcal{A}, \mathcal{B})$ of interactive algorithms over a classical channel. $\mathcal{A}(\mathcal{B})$ is a QPT algorithm that takes 1^λ as input and outputs a bit string a (b). We require the following properties.*

α -correctness:

$$\Pr[a = b : (a, b) \leftarrow \langle \mathcal{A}(1^\lambda), \mathcal{B}(1^\lambda) \rangle] \geq \alpha(\lambda). \quad (13)$$

Here, $(a, b) \leftarrow \langle \mathcal{A}(1^\lambda), \mathcal{B}(1^\lambda) \rangle$ means that $\mathcal{A}$'s output is a and $\mathcal{B}$'s output is b .

β -security: For any non-uniform QPT adversary $\mathcal{E}$,

$$\Pr[a = e : (a, b; \tau) \leftarrow \langle \mathcal{A}(1^\lambda), \mathcal{B}(1^\lambda) \rangle, e \leftarrow \mathcal{E}(\tau)] \leq \beta(\lambda). \quad (14)$$

Here, $(a, b; \tau) \leftarrow \langle \mathcal{A}(1^\lambda), \mathcal{B}(1^\lambda) \rangle$ means that $\mathcal{A}$'s output is a , $\mathcal{B}$'s output is b , and τ is the transcript.

If $(\mathcal{A}, \mathcal{B})$ is a $(1 - \text{negl}, \text{negl})$ -QCCC KE, then we simply say that $(\mathcal{A}, \mathcal{B})$ is a QCCC KE.

The following lemma was originally shown for classical KE, but we confirm that the proof also applies to QCCC KE.

Lemma 2.11 ([BLMP23], Lemma 2.13). *Let c_1 and c_2 be any constants such that $c_1 > c_2 > 0$. Let $p(\lambda) := \lambda^{c_1}$ and $q(\lambda) := \lambda^{c_2}$. If $(1 - \frac{1}{p}, 1 - \frac{1}{q})$ -QCCC KE exist, then QCCC KE exist.*

3 Proofs of Quantum Memory

In this section, we define proofs of quantum memory (PoQM). We also observe that PoQM generalize the notion of PoQ.

3.1 Definition

We first define PoQM as follows.

Definition 3.1 (Proofs of Quantum Memory (PoQM)). *Let $\alpha, \beta : \mathbb{N} \rightarrow [0, 1]$ be any functions. Let $m_1, m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any functions. An $(\alpha, \beta, m_1, m_2)$ -proof of quantum memory $((\alpha, \beta, m_1, m_2)$ -PoQM) is a set $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ of interactive algorithms over a classical channel with the following syntax.*

Initialization Phase: *In the initialization phase, $\mathcal{V}_1$ and $\mathcal{P}_1$ interact over a classical channel. $\mathcal{V}_1$ is a PPT algorithm that takes 1^λ as input and outputs a bit string v . $\mathcal{P}_1$ is a QPT algorithm that takes 1^λ as input and outputs a bit string state and an m_1 -qubit quantum state σ_{state} . In other words,*

$$(v, (\text{state}, \sigma_{\text{state}})) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle. \quad (15)$$

Execution Phase: In the execution phase, $\mathcal{V}_2$ and $\mathcal{P}_2$ interact over a classical channel. $\mathcal{V}_2$ is a PPT algorithm that takes v as input and outputs $\top$ or $\perp$. $\mathcal{P}_2$ is a QPT algorithm that takes state and σ_{state} as input and outputs nothing. In other words,

$$\top/\perp \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2(\text{state}, \sigma_{\text{state}}) \rangle. \quad (16)$$

We require the following two properties.

α -completeness: For all sufficiently large $\lambda \in \mathbb{N}$,

$$\Pr \left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2(\text{state}, \sigma_{\text{state}}) \rangle : (v, (\text{state}, \sigma_{\text{state}})) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle \right] \geq \alpha(\lambda). \quad (17)$$

(β, m_2) -soundness: For any non-uniform QPT adversary $\mathcal{P}_1^*$ that outputs a bit string s and an m_2 -qubit quantum state ρ , for any non-uniform ¹⁰ QPT adversary $\mathcal{P}_2^*$ that takes s and ρ as input, and for all sufficiently large $\lambda \in \mathbb{N}$,

$$\Pr \left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^*(s, \rho) \rangle : (v, (s, \rho)) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^*(1^\lambda) \rangle \right] \leq \beta(\lambda). \quad (18)$$

3.2 Amplification of m_2

We show that m_2 can be increased by increasing β .

Lemma 3.2. Let $\alpha, \beta : \mathbb{N} \rightarrow [0, 1]$ be any functions. Let $m_1, m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any functions. An $(\alpha, \beta, m_1, 0)$ -PoQM is an $(\alpha, 2^{m_2}\beta, m_1, m_2)$ -PoQM.

Proof of Lemma 3.2. Let $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ be an $(\alpha, \beta, m_1, 0)$ -PoQM. We show that this is also an $(\alpha, 2^{m_2}\beta, m_1, m_2)$ -PoQM. α -completeness is straightforward. For the sake of contradiction, we assume that the PoQM is not $(2^{m_2}\beta, m_2)$ -sound. This means that there exists an adversary $\mathcal{P}_1^{*(m_2)}$ that outputs an m_2 -qubit quantum state ρ and a bit string s , and an adversary $\mathcal{P}_2^{*(m_2)}$ that takes ρ and s as input such that

$$\Pr \left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^{*(m_2)}(s, \rho) \rangle : (v, (s, \rho)) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^{*(m_2)}(1^\lambda) \rangle \right] > 2^{m_2(\lambda)}\beta(\lambda) \quad (19)$$

for infinitely many $\lambda \in \mathbb{N}$.

From this $(\mathcal{P}_1^{*(m_2)}, \mathcal{P}_2^{*(m_2)})$, we can construct a pair $(\mathcal{P}_1^{*(0)}, \mathcal{P}_2^{*(0)})$ of adversaries that breaks $(\beta, 0)$ -soundness as follows.

- $\mathcal{P}_1^{*(0)}$: Run $(s, \rho) \leftarrow \mathcal{P}_1^{*(m_2)}(1^\lambda)$. Measure ρ in the computational basis to get a measurement result $p \in \{0, 1\}^{m_2(\lambda)}$. Output $s' := (s, p)$.
- $\mathcal{P}_2^{*(0)}$: Get $s' = (s, p)$ as input. Run $\mathcal{P}_2^{*(m_2)}(s, |p\rangle\langle p|)$.

By Lemma 2.1 and Equation (19),

$$\Pr \left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^{*(0)}(s') \rangle : (v, s') \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^{*(0)}(1^\lambda) \rangle \right] \quad (20)$$

$$\geq 2^{-m_2(\lambda)} \Pr \left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^{*(m_2)}(s, \rho) \rangle : (v, (s, \rho)) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^{*(m_2)}(1^\lambda) \rangle \right] \quad (21)$$

$$> \beta(\lambda) \quad (22)$$

for infinitely many $\lambda \in \mathbb{N}$. This contradicts $(\beta, 0)$ -soundness of the PoQM. $\square$

¹⁰In our setting, it is more natural that the non-uniform QPT adversary $\mathcal{P}_2^*$ takes only classical advice since we are interested in how much quantum memory the adversary can possess. However, we can construct PoQM with such stronger security, and therefore this only makes our results stronger.

3.3 Relation to PoQ

We can show that PoQ is a special case of PoQM with $m_2 = 0$.

Lemma 3.3. *Let $\alpha, \beta : \mathbb{N} \rightarrow [0, 1]$ be any functions. Let $m_1 : \mathbb{N} \rightarrow \mathbb{N}$ be any function. If $(\alpha, \beta, m_1, 0)$ -PoQM exist, then (α, β) -PoQ exist.*

Proof of Lemma 3.3. Assume that $(\alpha, \beta, m_1, 0)$ -PoQM exist. Let $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ be an $(\alpha, \beta, m_1, 0)$ -PoQM. From it, we construct an (α, β) -PoQ $(\mathcal{V}, \mathcal{P})$ as follows:

- $\top/\perp \leftarrow \langle \mathcal{V}(1^\lambda), \mathcal{P}(1^\lambda) \rangle$:
 1. $\mathcal{V}$ and $\mathcal{P}$ get 1^λ as input.
 2. $\mathcal{V}$ and $\mathcal{P}$ interact over a classical channel. $\mathcal{V}$ runs $v \leftarrow \mathcal{V}_1(1^\lambda)$, and $\mathcal{P}$ runs $(\text{state}, \sigma_{\text{state}}) \leftarrow \mathcal{P}_1(1^\lambda)$.
 3. $\mathcal{V}$ and $\mathcal{P}$ interact over a classical channel. $\mathcal{V}$ runs $\top/\perp \leftarrow \mathcal{V}_2(v)$, and $\mathcal{P}$ runs $\mathcal{P}_2(\text{state}, \sigma_{\text{state}})$. $\mathcal{V}$ outputs the output of $\mathcal{V}_2(v)$.

α -completeness of thus constructed PoQ is clear. Next we show β -soundness. For the sake of contradiction, we assume that the constructed PoQ is not β -sound. This means that there exists a non-uniform PPT prover $\mathcal{P}^*$ such that

$$\Pr[\top \leftarrow \langle \mathcal{V}(1^\lambda), \mathcal{P}^*(1^\lambda) \rangle] > \beta(\lambda) \quad (23)$$

for infinitely many $\lambda \in \mathbb{N}$. We divide $\mathcal{P}^*$ into two algorithms $\mathcal{P}_1^*$ and $\mathcal{P}_2^*$ such that $\mathcal{P}_1^*$ interacts with $\mathcal{V}_1$ and $\mathcal{P}_2^*$ interacts with $\mathcal{V}_2$. Because $\mathcal{P}^*$ is a PPT algorithm, $\mathcal{P}_1^*$ outputs only a classical bit string, which we call it s , and $\mathcal{P}_2^*$ takes only s as input. We can show that thus constructed $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ breaks $(\beta, 0)$ -soundness of the PoQM, because

$$\Pr[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^*(s) \rangle : (v, s) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^*(1^\lambda) \rangle] = \Pr[\top \leftarrow \langle \mathcal{V}(1^\lambda), \mathcal{P}^*(1^\lambda) \rangle] > \beta(\lambda) \quad (24)$$

for infinitely many $\lambda \in \mathbb{N}$. Hence the constructed PoQ is β -sound. $\square$

4 Constructions of PoQM

In this section, we provide two constructions of PoQM. The first construction is from 1-of- 2^k puzzles. The second one is from RSPs.

4.1 1-of- 2^k Puzzles Imply PoQM

Theorem 4.1. *Let $m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any polynomially bounded function. Let $c > 0$ be any constant such that $m_2(\lambda) = O(\lambda^c)$. Let k be any polynomial such that $k(\lambda) = \omega(\lambda^c)$. If 1-of- 2^k puzzles with c -soundness exist, then, 4-round $(1 - \text{negl}, \text{negl}, m_1, m_2)$ -PoQM exist with some polynomial m_1 .*

By combining this theorem with Lemma 2.7, we obtain the following corollary.

Corollary 4.2. *Let $m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any polynomially bounded function. Assuming the subexponential hardness of LWE, 4-round $(1 - \text{negl}, \text{negl}, m_1, m_2)$ -PoQM exist with some polynomial m_1 .*

Proof of Theorem 4.1. Assume that 1-of- 2^k puzzles with c -soundness exist. Let $(\text{KeyGen}, \text{Obligate}, \text{Solve}, \text{Ver})$ be a 1-of- 2^k puzzle with c -soundness. We construct a PoQM $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ as follows:

Initialization Phase:

1. $\mathcal{V}_1$ and $\mathcal{P}_1$ get 1^λ as input.
2. $\mathcal{V}_1$ runs $(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$ and sends pk to $\mathcal{P}_1$.
3. $\mathcal{P}_1$ runs $(y, \rho) \leftarrow \text{Obligate}(pk)$ and sends y to $\mathcal{V}_1$. The number of qubits of ρ is $m_1(\lambda)$.
4. $\mathcal{V}_1$ outputs $v := (sk, y)$. $\mathcal{P}_1$ outputs $(\text{state}, \sigma_{\text{state}}) := ((pk, y), \rho)$.

Execution Phase:

1. $\mathcal{V}_2$ takes v as input. $\mathcal{P}_2$ takes $(\text{state}, \sigma_{\text{state}})$ as input.
2. $\mathcal{V}_2$ samples $ch \leftarrow \{0, 1\}^{k(\lambda)}$ and sends it to $\mathcal{P}_2$.
3. $\mathcal{P}_2$ runs $\text{ans} \leftarrow \text{Solve}(pk, y, \rho, ch)$ and sends ans to $\mathcal{V}_2$.
4. $\mathcal{V}_2$ runs $\top/\perp \leftarrow \text{Ver}(sk, y, ch, \text{ans})$ and outputs its output.

Our goal is to show that the constructed $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ is a $(1 - \text{negl}, \text{negl}, m_1, m_2)$ -PoQM. We achieve this goal with the following three steps:

1. We show that $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ is a $(1 - \text{negl}, \epsilon, m_1, 0)$ -PoQM, where $\epsilon(\lambda) := (2^{-k(\lambda)} + \text{negl}(2^{\lambda^c}))^{\frac{1}{2}}$.
2. Using Lemma 3.2, a $(1 - \text{negl}, \epsilon, m_1, 0)$ -PoQM is a $(1 - \text{negl}, 2^{m_2} \epsilon, m_1, m_2)$ -PoQM.
3. We show that $2^{m_2(\lambda)} \epsilon(\lambda) = \text{negl}(\lambda)$.

The second step is straightforward. In the following, we will explain the first and third steps.

First step. $(1 - \text{negl})$ -completeness is straightforward. Let us show $(\epsilon, 0)$ -soundness. For the sake of contradiction, assume that it is not $(\epsilon, 0)$ -sound. Then there exists a pair $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ of adversaries such that

$$\epsilon(\lambda) < \Pr \left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^*(s) \rangle : (v, s) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^*(1^\lambda) \rangle \right] \quad (25)$$

$$= \Pr \left[\top \leftarrow \text{Ver}(sk, y, ch, \text{ans}) : \begin{array}{l} (pk, sk) \leftarrow \text{KeyGen}(1^\lambda) \\ (y, s') \leftarrow \mathcal{P}_1^*(pk) \\ ch \leftarrow \{0, 1\}^{k(\lambda)} \\ \text{ans} \leftarrow \mathcal{P}_2^*(ch, s') \end{array} \right] \quad (26)$$

for infinitely many $\lambda \in \mathbb{N}$. From this $(\mathcal{P}_1^*, \mathcal{P}_2^*)$, we can construct a set $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ of adversaries that breaks c -soundness of the 1-of- 2^k puzzle as follows:

- $\mathcal{A}$: Run $(y, s') \leftarrow \mathcal{P}_1^*(pk)$. Send y to Chal and send s' to $\mathcal{B}$ and $\mathcal{C}$.
- $\mathcal{B}$: Run $\text{ans}_{\mathcal{B}} \leftarrow \mathcal{P}_2^*(ch, s')$ and send $\text{ans}_{\mathcal{B}}$ to Chal.
- $\mathcal{C}$: Run $\text{ans}_{\mathcal{C}} \leftarrow \mathcal{P}_2^*(ch, s')$ and send $\text{ans}_{\mathcal{C}}$ to Chal.

$(\mathcal{A}, \mathcal{B}, \mathcal{C})$ can break c -soundness as follows:

$$\Pr[\top \leftarrow \text{Chal}] = \Pr \left[\begin{array}{l} (\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda) \\ (y, s') \leftarrow \mathcal{P}_1^*(\text{pk}) \\ \text{ch} \leftarrow \{0, 1\}^{k(\lambda)} \\ \text{ans}_{\mathcal{B}} \leftarrow \mathcal{P}_2^*(\text{ch}, s') \\ \text{ans}_{\mathcal{C}} \leftarrow \mathcal{P}_2^*(\text{ch}, s') \end{array} : \top \leftarrow \text{Ver}(\text{sk}, y, \text{ch}, \text{ans}_{\mathcal{B}}) \wedge \top \leftarrow \text{Ver}(\text{sk}, y, \text{ch}, \text{ans}_{\mathcal{C}}) \right] \quad (27)$$

$$\geq \Pr \left[\begin{array}{l} (\text{pk}, \text{sk}) \leftarrow \text{KeyGen}(1^\lambda) \\ (y, s') \leftarrow \mathcal{P}_1^*(\text{pk}) \\ \text{ch} \leftarrow \{0, 1\}^{k(\lambda)} \\ \text{ans} \leftarrow \mathcal{P}_2^*(\text{ch}, s') \end{array} : \top \leftarrow \text{Ver}(\text{sk}, y, \text{ch}, \text{ans}) \right]^2 \quad (28)$$

$$> \epsilon(\lambda)^2 = 2^{-k(\lambda)} + \text{negl}(2^{\lambda^c}) \quad (29)$$

for infinitely many $\lambda \in \mathbb{N}$. The first inequality follows from the Jensen's inequality. Hence, $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ break c -soundness of the 1-of- 2^k puzzle, contradicting the assumption.

Third step. Since $k(\lambda) = \omega(\lambda^c)$, we have $2^{-k(\lambda)} = \text{negl}(2^{\lambda^c})$, and thus $\epsilon(\lambda) = (2^{-k(\lambda)} + \text{negl}(2^{\lambda^c}))^{\frac{1}{2}} = \text{negl}(2^{\lambda^c})$. From $m_2(\lambda) = O(\lambda^c)$, there exists a polynomial p such that $2^{m_2(\lambda)} \leq p(2^{\lambda^c})$ for all sufficiently large $\lambda \in \mathbb{N}$. Hence, we obtain

$$2^{m_2(\lambda)} \epsilon(\lambda) \leq p(2^{\lambda^c}) \text{negl}(2^{\lambda^c}) = \text{negl}(2^{\lambda^c}) = \text{negl}(\lambda). \quad (30)$$

□

4.2 RSPs Imply PoQM

Theorem 4.3. *Let p be any polynomial. Let $m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any polynomially bounded function such that $m_2(\lambda) = \omega(\log(\lambda))$. If $(\lceil 9.1m_2 \rceil, \frac{1}{2p})$ -RSPs exist, then $(1 - \text{negl}, 1/p, \lceil 9.1m_2 \rceil, m_2)$ -PoQM exist.*

By combining this theorem with Lemma 2.9, we obtain the following corollary.

Corollary 4.4. *Let p be any polynomial. Let $m_2 : \mathbb{N} \rightarrow \mathbb{N}$ be any polynomially bounded function such that $m_2(\lambda) = \omega(\log(\lambda))$. Assuming the polynomial hardness of LWE, r -round $(1 - \text{negl}, 1/p, \lceil 9.1m_2 \rceil, m_2)$ -PoQM exist with a certain polynomial r .*

Proof of Theorem 4.3. Assume that $(\lceil 9.1m_2 \rceil, \frac{1}{2p})$ -RSPs exist. Let $(\mathcal{V}, \mathcal{P})$ be a $(\lceil 9.1m_2 \rceil, \frac{1}{2p})$ -RSP. We construct a $(1 - \text{negl}, 1/p, \lceil 9.1m_2 \rceil, m_2)$ -PoQM $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ as follows:

Initialization Phase:

1. $\mathcal{V}_1$ and $\mathcal{P}_1$ take 1^λ as input.
2. $\mathcal{V}_1$ runs $v \leftarrow \mathcal{V}(1^\lambda)$ where $v \in \{(\text{pass}, x, \theta), \text{fail}\}$ and $\mathcal{P}_1$ runs $\phi \leftarrow \mathcal{P}(1^\lambda)$. Here, x and θ are $\lceil 9.1m_2 \rceil$ -bit strings, and ϕ is a $\lceil 9.1m_2 \rceil$ -qubit state.
3. $\mathcal{V}_1$'s output is v . $\mathcal{P}_1$'s output is $(\text{state}, \sigma_{\text{state}}) := (1^\lambda, \phi)$.

Execution Phase:

1. $\mathcal{V}_2$ takes $v \in \{(\text{pass}, x, \theta), \text{fail}\}$ as input. $\mathcal{P}_2$ takes $(\text{state}, \sigma_{\text{state}}) = (1^\lambda, \phi)$ as input.
2. If $v = \text{fail}$, then $\mathcal{V}_2$ samples $\theta \leftarrow \{0, 1\}^{\lceil 9.1m_2 \rceil}$ and sends it to $\mathcal{P}_2$. If $v = (\text{pass}, x, \theta)$, $\mathcal{V}_2$ sends θ to $\mathcal{P}_2$.
3. For each $i \in [\lceil 9.1m_2 \rceil]$, $\mathcal{P}_2$ measures i th qubit of ϕ in the computational basis if $\theta_i = 0$ or in the Hadamard basis if $\theta_i = 1$. Let x'_i be the measurement result on the i th qubit. $\mathcal{P}_2$ sets $x' := x'_1 \parallel \dots \parallel x'_{\lceil 9.1m_2 \rceil}$.
4. $\mathcal{P}_2$ sends x' to $\mathcal{V}_2$.
5. If $v = \text{fail}$ or $x \neq x'$, $\mathcal{V}_2$ outputs $\perp$. Otherwise, $\mathcal{V}_2$ outputs $\top$.

Now we show that the constructed $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ is a $(1 - \text{negl}, 1/p, \lceil 9.1m_2 \rceil, m_2)$ -PoQM.

$(1 - \text{negl})$ -completeness is straightforward. Let us next show $(1/p, m_2)$ -soundness. We define **Hybrid**₀ as follows, which is the original security game for $(1/p, m_2)$ -soundness.

Hybrid₀:

1. $\mathcal{V}_1$ and $\mathcal{P}_1^*$ take 1^λ as input.
2. They run the RSP. $\mathcal{V}_1$ outputs $v \in \{(\text{pass}, x, \theta), \text{fail}\}$. $\mathcal{P}_1^*$ outputs a quantum state $\sigma_{\mathbf{Q}'}$ on the register $\mathbf{Q}'$.
3. $\mathcal{P}_1^*$ runs a certain QPT algorithm E on $\sigma_{\mathbf{Q}'}$ to get (s, ρ) , where s is a classical bit string and ρ is an m_2 -qubit quantum state: $(s, \rho) \leftarrow E(\sigma_{\mathbf{Q}'})$. $\mathcal{P}_1^*$ outputs (s, ρ) .
4. $\mathcal{V}_2$ takes v as input. $\mathcal{P}_2^*$ takes (s, ρ) as input.
5. If $v = \text{fail}$, then $\mathcal{V}_2$ samples $\theta \leftarrow \{0, 1\}^{\lceil 9.1m_2 \rceil}$ and sends it to $\mathcal{P}_2^*$. If $v = (\text{pass}, x, \theta)$, $\mathcal{V}_2$ sends θ to $\mathcal{P}_2^*$.
6. $\mathcal{P}_2^*$ sends x' to $\mathcal{V}_2$.
7. If $v = \text{fail}$ or $x \neq x'$, $\mathcal{V}_2$ outputs $\perp$. Otherwise, $\mathcal{V}_2$ outputs $\top$.

Because of $\frac{1}{2p}$ -soundness of the RSP (Equation (12)), there exists a non-uniform QPT algorithm Sim . By using it, we define **Hybrid**₁, which is the same as **Hybrid**₀ except for the step 2:

Hybrid₁:

2. $\mathcal{V}_1$ samples $(x, \theta) \leftarrow \{0, 1\}^{\lceil 9.1m_2 \rceil} \times \{0, 1\}^{\lceil 9.1m_2 \rceil}$ and generates $\text{Sim}(\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i})$, which consists of two registers $\mathbf{F}$ and $\mathbf{Q}'$. $\mathcal{V}_1$ gets $\text{flag} \in \{\text{pass}, \text{fail}\}$ by measuring the register $\mathbf{F}$ and sends the register $\mathbf{Q}'$ of the post-measurement state to $\mathcal{P}_1^*$. $\mathcal{V}_1$ sets $v := \text{fail}$ if $\text{flag} = \text{fail}$. Otherwise, $\mathcal{V}_1$ sets $v := (\text{pass}, x, \theta)$.

Lemma 4.5.

$$|\Pr[\top \leftarrow \mathbf{Hybrid}_0(\lambda)] - \Pr[\top \leftarrow \mathbf{Hybrid}_1(\lambda)]| \leq \frac{1}{2p(\lambda)} \quad (31)$$

for all sufficiently large $\lambda \in \mathbb{N}$.

Proof of Lemma 4.5. For the sake of contradiction, we assume that

$$|\Pr[\top \leftarrow \mathbf{Hybrid}_0(\lambda)] - \Pr[\top \leftarrow \mathbf{Hybrid}_1(\lambda)]| > \frac{1}{2p(\lambda)} \quad (32)$$

for infinitely many $\lambda \in \mathbb{N}$. Then, we can construct a non-uniform QPT algorithm $\mathcal{D}$ that breaks $\frac{1}{2p}$ -soundness of $(\lceil 9.1m_2 \rceil, \frac{1}{2p})$ -RSP as follows:

1. Get a quantum state over registers $\mathbf{F}, \mathbf{D}, \mathbf{Q}'$, where registers are defined as in Definition 2.8.
2. Get $(x, \theta) \in \{0, 1\}^{\lceil 9.1m_2 \rceil} \times \{0, 1\}^{\lceil 9.1m_2 \rceil}$ by measuring register $\mathbf{D}$. Set $v = (\text{pass}, x, \theta)$.
3. Run $(s, \rho) \leftarrow E(\xi_{\mathbf{Q}'})$, where E is the algorithm of step 3 of $\mathbf{Hybrid}_0$ and $\xi_{\mathbf{Q}'}$ is the reduced state on the register $\mathbf{Q}'$ of the post-measurement state.
4. Simulate the interaction between $\mathcal{V}_2$ and $\mathcal{P}_2^*$ from the step 4 of $\mathbf{Hybrid}_0$ to the last step.
5. Output $\top$ if $\mathcal{V}_2$ outputs $\top$. Otherwise, output $\perp$.

It is clear that

$$\Pr[\top \leftarrow \mathbf{Hybrid}_0(\lambda)] = \text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'}] \Pr\left[\top \leftarrow \mathcal{D}\left(\frac{\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'} \Pi_{\mathbf{F}}^{\text{pass}}}{\text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'}]}\right)\right], \quad (33)$$

where $\sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'}$ is the output of $\mathcal{V}_1$ and $\mathcal{P}_1^*$ at step 2 of $\mathbf{Hybrid}_0$, and

$$\Pr[\top \leftarrow \mathbf{Hybrid}_1(\lambda)] = \text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}})] \Pr\left[\top \leftarrow \mathcal{D}\left(\frac{\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}}) \Pi_{\mathbf{F}}^{\text{pass}}}{\text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}})]}\right)\right]. \quad (34)$$

By Equation (32),

$$\left| \text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'}] \Pr\left[\top \leftarrow \mathcal{D}\left(\frac{\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'} \Pi_{\mathbf{F}}^{\text{pass}}}{\text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \sigma_{\mathbf{F}, \mathbf{D}, \mathbf{Q}'}]}\right)\right] \right. \quad (35)$$

$$\left. - \text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}})] \Pr\left[\top \leftarrow \mathcal{D}\left(\frac{\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}}) \Pi_{\mathbf{F}}^{\text{pass}}}{\text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\eta_{\mathbf{D}, \mathbf{Q}})]}\right)\right] \right| \quad (36)$$

$$= |\Pr[\top \leftarrow \mathbf{Hybrid}_0(\lambda)] - \Pr[\top \leftarrow \mathbf{Hybrid}_1(\lambda)]| > \frac{1}{2p(\lambda)} \quad (37)$$

for infinitely many $\lambda \in \mathbb{N}$. This contradicts $\frac{1}{2p}$ -soundness of the RSP. $\square$

Let us define $\mathbf{Hybrid}_2$, which is the same as $\mathbf{Hybrid}_1$ except for the step 2:

Hybrid₂:

2. $\mathcal{V}_1$ samples $(x, \theta) \leftarrow \{0, 1\}^{\lceil 9.1m_2 \rceil} \times \{0, 1\}^{\lceil 9.1m_2 \rceil}$, generates $\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i}$ and sends it to $\mathcal{P}_1^*$. $\mathcal{V}_1$ sets $v := (\text{pass}, x, \theta)$

As shown below, the acceptance probability of $\mathbf{Hybrid}_2$ is at least that of $\mathbf{Hybrid}_1$.

Lemma 4.6.

$$\Pr[\top \leftarrow \mathbf{Hybrid}_1(\lambda)] \leq \Pr[\top \leftarrow \mathbf{Hybrid}_2(\lambda)] \quad (38)$$

for all $\lambda \in \mathbb{N}$.

Proof of Lemma 4.6. We start by expanding the acceptance probability of $\mathbf{Hybrid}_1$.

$$\Pr[\top \leftarrow \mathbf{Hybrid}_1(\lambda)] \quad (39)$$

$$= \Pr[\text{pass}] \Pr \left[\top \leftarrow \langle \mathcal{V}_2(\text{pass}, x, \theta), \mathcal{P}_2^*(s, \rho) \rangle : \begin{array}{l} (x, \theta) \leftarrow \{0, 1\}^{\lceil 9.1m_2 \rceil} \times \{0, 1\}^{\lceil 9.1m_2 \rceil} \\ (s, \rho) \leftarrow E(\zeta_{\mathbf{Q}'}^{x, \theta}) \end{array} \right]. \quad (40)$$

Here, $\zeta_{\mathbf{Q}'}^{x, \theta}$ is the reduced state on register $\mathbf{Q}'$ of

$$\frac{\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i}) \Pi_{\mathbf{F}}^{\text{pass}}}{\text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i})]}, \quad (41)$$

and

$$\Pr[\text{pass}] = \text{Tr} \left[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim} \left(\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i} \right) \right]. \quad (42)$$

Next, we write down explicitly the acceptance probability of $\mathbf{Hybrid}_2$. Then we bound this probability by considering the special case where $\mathcal{P}_1^*$, as its first step after applying Sim , measures register $\mathbf{F}$, and condition on the measurement outcome being pass. We further restrict by replacing the state with its reduced version. Since these restrictions can only reduce the acceptance probability, the resulting experiment provides a lowerbound for $\mathbf{Hybrid}_2$, which coincides exactly with the acceptance probability of $\mathbf{Hybrid}_1$.

$$\Pr[\top \leftarrow \mathbf{Hybrid}_2(\lambda)] \quad (43)$$

$$= \Pr \left[\top \leftarrow \langle \mathcal{V}_2(\text{pass}, x, \theta), \mathcal{P}_2^*(s, \rho) \rangle : \begin{array}{l} (x, \theta) \leftarrow \{0, 1\}^{\lceil 9.1m_2 \rceil} \times \{0, 1\}^{\lceil 9.1m_2 \rceil} \\ (s, \rho) \leftarrow E \left(\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i} \right) \end{array} \right] \quad (44)$$

$$\geq \Pr[\text{pass}] \Pr \left[\top \leftarrow \langle \mathcal{V}_2(\text{pass}, x, \theta), \mathcal{P}_2^*(s, \rho) \rangle : \begin{array}{l} (x, \theta) \leftarrow \{0, 1\}^{\lceil 9.1m_2 \rceil} \times \{0, 1\}^{\lceil 9.1m_2 \rceil} \\ (s, \rho) \leftarrow E \left(\frac{\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i}) \Pi_{\mathbf{F}}^{\text{pass}}}{\text{Tr}[\Pi_{\mathbf{F}}^{\text{pass}} \text{Sim}(\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle \langle x_i| H^{\theta_i})]} \right) \end{array} \right] \quad (45)$$

$$\geq \Pr[\text{pass}] \Pr \left[\top \leftarrow \langle \mathcal{V}_2(\text{pass}, x, \theta), \mathcal{P}_2^*(s, \rho) \rangle : \begin{array}{l} (x, \theta) \leftarrow \{0, 1\}^{\lceil 9.1m_2 \rceil} \times \{0, 1\}^{\lceil 9.1m_2 \rceil} \\ (s, \rho) \leftarrow E(\zeta_{\mathbf{Q}'}^{x, \theta}) \end{array} \right] \quad (46)$$

$$= \Pr[\top \leftarrow \mathbf{Hybrid}_1(\lambda)]. \quad (47)$$

□

We define $\mathbf{Hybrid}_3$, which is the same as $\mathbf{Hybrid}_2$ except for steps 3 and 4:

Hybrid₃:

3. $\mathcal{P}_1^*$ runs a certain QPT algorithm E on $\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle\langle x_i| H^{\theta_i}$ to get (s, ρ) , where s is a classical bit string and ρ is an m_2 -qubit quantum state: $(s, \rho) \leftarrow E(\bigotimes_{i=1}^{\lceil 9.1m_2 \rceil} H^{\theta_i} |x_i\rangle\langle x_i| H^{\theta_i})$. Get a bit string $p \in \{0, 1\}^{m_2}$ by measuring ρ in the computational basis. Set $s' := (s, p)$. $\mathcal{P}_1^*$ outputs s' .
4. $\mathcal{V}_2$ takes v as input. $\mathcal{P}_2^*$ takes $s' = (s, p)$ as input, and uses it as $(s, |p\rangle\langle p|)$.

By Lemma 2.1, we can obtain the following lemma.

Lemma 4.7.

$$\Pr[\top \leftarrow \mathbf{Hybrid}_2(\lambda)] \leq 2^{m_2(\lambda)} \Pr[\top \leftarrow \mathbf{Hybrid}_3(\lambda)] \quad (48)$$

for all $\lambda \in \mathbb{N}$.

To conclude the theorem, we show the following lemma.

Lemma 4.8. For all sufficiently large $\lambda \in \mathbb{N}$,

$$\Pr[\top \leftarrow \mathbf{Hybrid}_3(\lambda)] \leq 2^{-\frac{\xi}{2} \cdot \lceil 9.1m_2(\lambda) \rceil + 2^{-\lceil 9.1m_2(\lambda) \rceil}}. \quad (49)$$

Here, $\xi = -\log\left(\frac{1}{2} + \frac{1}{2\sqrt{2}}\right) > 0.22$.

Proof of Lemma 4.8. For the sake of contradiction, we assume that there exists a pair $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ of adversaries such that

$$2^{-\frac{\xi}{2} \cdot \lceil 9.1m_2(\lambda) \rceil + 2^{-\lceil 9.1m_2(\lambda) \rceil}} < \Pr[\top \leftarrow \mathbf{Hybrid}_3(\lambda)] \quad (50)$$

$$= \Pr \left[\begin{array}{l} (x, \theta) \leftarrow \{0, 1\}^{\lceil 9.1m_2(\lambda) \rceil} \times \{0, 1\}^{\lceil 9.1m_2(\lambda) \rceil} \\ x = x' : \begin{array}{l} (s, p) \leftarrow \mathcal{P}_1^*(\bigotimes_{i=1}^{\lceil 9.1m_2(\lambda) \rceil} H^{\theta_i} |x_i\rangle\langle x_i| H^{\theta_i}) \\ x' \leftarrow \mathcal{P}_2^*(\theta, s, |p\rangle\langle p|) \end{array} \end{array} \right] \quad (51)$$

for infinitely many $\lambda \in \mathbb{N}$. From this $(\mathcal{P}_1^*, \mathcal{P}_2^*)$, we can construct a non-uniform QPT adversary $\mathcal{A}$ that breaks Lemma 2.2 as follows:

1. Send the classical description of $\mathcal{P}_1^*$ to $\mathcal{C}$.
2. $\mathcal{C}$ runs $(s, p) \leftarrow \mathcal{P}_1^*(\bigotimes_{i=1}^{\lceil 9.1m_2(\lambda) \rceil} H^{\theta_i} |x_i\rangle\langle x_i| H^{\theta_i})$ and returns (s, p) to $\mathcal{A}$.
3. Receive θ from $\mathcal{C}$, run $x' \leftarrow \mathcal{P}_2^*(\theta, s, |p\rangle\langle p|)$ and send x' to $\mathcal{C}$.

Then, for infinitely many $\lambda \in \mathbb{N}$,

$$\Pr[\top \leftarrow \mathcal{C}] = \Pr \left[\begin{array}{l} (x, \theta) \leftarrow \{0, 1\}^{\lceil 9.1m_2(\lambda) \rceil} \times \{0, 1\}^{\lceil 9.1m_2(\lambda) \rceil} \\ x = x' : \begin{array}{l} (s, p) \leftarrow \mathcal{P}_1^*(\bigotimes_{i=1}^{\lceil 9.1m_2(\lambda) \rceil} H^{\theta_i} |x_i\rangle\langle x_i| H^{\theta_i}) \\ x' \leftarrow \mathcal{P}_2^*(\theta, s, |p\rangle\langle p|) \end{array} \end{array} \right] \quad (52)$$

$$= \Pr[\top \leftarrow \mathbf{Hybrid}_3(\lambda)] \quad (53)$$

$$> 2^{-\frac{\xi}{2} \cdot \lceil 9.1m_2(\lambda) \rceil + 2^{-\lceil 9.1m_2(\lambda) \rceil}}. \quad (54)$$

This contradicts Lemma 2.2. □

By combining Lemmata 4.5 to 4.8, we have $\Pr[\top \leftarrow \mathbf{Hybrid}_0(\lambda)] < 1/2p(\lambda) + \text{negl}(\lambda) < 1/p(\lambda)$ for all sufficiently large $\lambda \in \mathbb{N}$. □

5 Lowerbounds of PoQM

In this section, we show that PoQM imply StatePuzzs, and extractable PoQM imply QCCC KE.

5.1 PoQM imply StatePuzzs

We first show that PoQM imply StatePuzzs.

Theorem 5.1. *Let $\alpha, \beta : \mathbb{N} \rightarrow [0, 1]$ be any functions such that $\alpha(\lambda) - \beta(\lambda) \geq 1/\text{poly}(\lambda)$ for all sufficiently large $\lambda \in \mathbb{N}$. Let $m_1 : \mathbb{N} \rightarrow \mathbb{N}$ be any function. If $(\alpha, \beta, m_1, 0)$ -PoQM exist, then StatePuzzs exist.*

Proof of Theorem 5.1. Assume that $(\alpha, \beta, m_1, 0)$ -PoQM exist. Let $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ be an $(\alpha, \beta, m_1, 0)$ -PoQM. The final state before the measurement of $\mathcal{P}_1$ is written as $\sum_{\text{state}} c_{\text{state}} |\phi_{\text{state}}\rangle |\text{state}\rangle$ with some complex coefficients $\{c_{\text{state}}\}$, where $|\phi_{\text{state}}\rangle$ is a pure m'_1 -qubit state. $\mathcal{P}_1$ measures the second register to get the result state. $\mathcal{P}_1$ outputs $(\text{state}, \sigma_{\text{state}})$, where σ_{state} is the first m_1 qubits of $|\phi_{\text{state}}\rangle$.

From $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$, we construct an $(\alpha, \beta, m'_1, 0)$ -PoQM $(\mathcal{V}_1, \mathcal{P}'_1, \mathcal{V}_2, \mathcal{P}'_2)$ as follows:

1. $\mathcal{P}'_1$ generates $\sum_{\text{state}} c_{\text{state}} |\phi_{\text{state}}\rangle |\text{state}\rangle$, measures the second register, and outputs $(\text{state}, |\phi_{\text{state}}\rangle)$.
2. $\mathcal{P}'_2$ takes $(\text{state}, |\phi_{\text{state}}\rangle)$ as input, and runs $\mathcal{P}_2(\text{state}, \sigma_{\text{state}})$, where σ_{state} is the first m_1 qubits of $|\phi_{\text{state}}\rangle$.

From $(\mathcal{V}_1, \mathcal{P}'_1, \mathcal{V}_2, \mathcal{P}'_2)$, we construct a StatePuzz, Samp, as follows:

1. Take 1^λ as input.
2. Run $(v, (\text{state}, |\phi_{\text{state}}\rangle)) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}'_1(1^\lambda) \rangle$. Let τ be the transcript.
3. Output $s := (\text{state}, \tau)$ and $|\psi_s\rangle := |\phi_{\text{state}}\rangle$.

Now we show that thus constructed Samp is a $1/p$ -StatePuzz with a certain polynomial p . From Lemma 2.5, such a $1/p$ -StatePuzz can be amplified to obtain a StatePuzz.

Let p be a polynomial such that $p(\lambda) > (\alpha(\lambda) - \beta(\lambda))^{-2}$ for all sufficiently large $\lambda \in \mathbb{N}$. For the sake of contradiction, we assume that Samp is not a $1/p$ -StatePuzz. Then there exists a non-uniform QPT algorithm $\mathcal{A}$ such that for infinitely many $\lambda \in \mathbb{N}$,

$$\mathbb{E}_{(s, |\psi_s\rangle) \leftarrow \text{Samp}(1^\lambda)} \langle \psi_s | \mathcal{A}(s) | \psi_s \rangle > 1 - \frac{1}{p(\lambda)}. \quad (55)$$

From this $\mathcal{A}$, we construct a pair $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ of adversaries that breaks $(\beta, 0)$ -soundness of the PoQM as follows:

- $\mathcal{P}_1^*$: Run $(\text{state}, |\phi_{\text{state}}\rangle) \leftarrow \mathcal{P}'_1(1^\lambda)$. Let τ be the transcript. Output (state, τ) .
- $\mathcal{P}_2^*$: Run $\mathcal{A}(\text{state}, \tau)$. Run $\mathcal{P}'_2(\text{state}, \mathcal{A}(\text{state}, \tau))$.

$(\mathcal{P}_1^*, \mathcal{P}_2^*)$ can break $(\beta, 0)$ -soundness of the PoQM as follows.

$$\Pr\left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^*(\text{state}, \tau) \rangle : (v, (\text{state}, \tau)) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^*(1^\lambda) \rangle\right] \quad (56)$$

$$= \Pr\left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2'(\text{state}, \mathcal{A}(\text{state}, \tau)) \rangle : (v, (\text{state}, |\phi_{\text{state}}\rangle)) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1'(1^\lambda) \rangle\right] \quad (57)$$

$$\geq \Pr\left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2'(\text{state}, |\phi_{\text{state}}\rangle) \rangle : (v, (\text{state}, |\phi_{\text{state}}\rangle)) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1'(1^\lambda) \rangle\right] \quad (58)$$

$$- \mathbb{E}_{((\text{state}, \tau), |\phi_{\text{state}}\rangle) \leftarrow \text{Samp}(1^\lambda)} \text{TD}(|\phi_{\text{state}}\rangle, \mathcal{A}(\text{state}, \tau)) \quad (59)$$

$$\geq \alpha(\lambda) - \mathbb{E}_{((\text{state}, \tau), |\phi_{\text{state}}\rangle) \leftarrow \text{Samp}(1^\lambda)} \sqrt{1 - \langle \phi_{\text{state}} | \mathcal{A}(\text{state}, \tau) | \phi_{\text{state}} \rangle} \quad (60)$$

for all sufficiently large $\lambda \in \mathbb{N}$. By Jensen's inequality and Equation (55),

$$\mathbb{E}_{((\text{state}, \tau), |\phi_{\text{state}}\rangle) \leftarrow \text{Samp}(1^\lambda)} \sqrt{1 - \langle \phi_{\text{state}} | \mathcal{A}(\text{state}, \tau) | \phi_{\text{state}} \rangle} \leq \sqrt{1 - \mathbb{E}_{((\text{state}, \tau), |\phi_{\text{state}}\rangle) \leftarrow \text{Samp}(1^\lambda)} \langle \phi_{\text{state}} | \mathcal{A}(\text{state}, \tau) | \phi_{\text{state}} \rangle} \quad (61)$$

$$< \frac{1}{p(\lambda)^{\frac{1}{2}}} \quad (62)$$

for infinitely many $\lambda \in \mathbb{N}$. Therefore,

$$\Pr\left[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^*(\text{state}, \tau) \rangle : (v, (\text{state}, \tau)) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^*(1^\lambda) \rangle\right] > \alpha(\lambda) - \frac{1}{p(\lambda)^{\frac{1}{2}}} > \beta(\lambda) \quad (63)$$

for infinitely many $\lambda \in \mathbb{N}$. This contradicts $(\beta, 0)$ -soundness of the PoQM. $\square$

5.2 Extractable PoQM Imply QCCC KE

We next show that a restricted version of PoQM, which we call extractable PoQM, implies QCCC KE.

Extractable PoQM are defined as follows.

Definition 5.2 (Extractable PoQM). Let $\gamma : \mathbb{N} \rightarrow [0, 1]$ be any function. We call an $(\alpha, \beta, m_1, m_2)$ -PoQM an $(\alpha, \beta, m_1, m_2)$ -extractable PoQM with extraction probability γ if the execution phase is the following.

Execution Phase: In the execution phase, the interaction is of a single round (i.e., of two-message):

1. $\mathcal{V}_2$ takes v as input.
2. $\mathcal{P}_2$ takes $(\text{state}, \sigma_{\text{state}})$ as input.
3. $\mathcal{V}_2$ sends a bit string x to $\mathcal{P}_2$.
4. $\mathcal{P}_2$ sends a bit string y to $\mathcal{V}_2$.
5. $\mathcal{V}_2$ outputs $\top$ or $\perp$.

Moreover, we require that there exists a QPT algorithm Ext such that

$$\Pr\left[y \leftarrow \text{Ext}(v, \tau, x) : \begin{array}{l} (v, (\text{state}, \sigma_{\text{state}}); \tau) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle \\ x \leftarrow \mathcal{V}_2(v) \\ y \leftarrow \mathcal{P}_2(\text{state}, \sigma_{\text{state}}, x) \end{array}\right] \geq \gamma(\lambda). \quad (64)$$

Here, $(v, (\text{state}, \sigma_{\text{state}}); \tau) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle$ means that $\mathcal{V}_1$'s output is v , $\mathcal{P}_1$'s output is $(\text{state}, \sigma_{\text{state}})$, and τ is the transcript.

The construction of PoQM from RSPs in Section 4.2 realizes the extractable PoQM. Thus, we obtain the following lemma.

Lemma 5.3. *Let p be any polynomial. Let m_2 be any polynomially bounded function such that $m_2(\lambda) = \omega(\log(\lambda))$. Assuming the polynomial hardness of LWE, r -round $(1 - \text{negl}, 1/p, \lceil 9.1m_2 \rceil, m_2)$ -extractable PoQM with extraction probability $1 - \text{negl}$ exist with a certain polynomial r .*

We show that extractable PoQM imply QCCC KE.

Theorem 5.4. *Let $m_1 : \mathbb{N} \rightarrow \mathbb{N}$ be any function. Let $\alpha : \mathbb{N} \rightarrow [0, 1]$ be any function. Let c_1 and c_2 be any constants such that $c_1 > c_2 > 0$. Let $p(\lambda) := \lambda^{c_1}$ and $q(\lambda) := \lambda^{c_2}$. If $(\alpha, \alpha - \frac{1}{q}, m_1, 0)$ -extractable PoQM with extraction probability $1 - \frac{1}{p}$ exist, then QCCC KE exist.*

Proof of Theorem 5.4. Assume that $(\alpha, \alpha - \frac{1}{q}, m_1, 0)$ -extractable PoQM with extraction probability $1 - \frac{1}{p}$ exist. Let $(\mathcal{V}_1, \mathcal{P}_1, \mathcal{V}_2, \mathcal{P}_2)$ be an $(\alpha, \alpha - \frac{1}{q}, m_1, 0)$ -extractable PoQM with extraction probability $1 - \frac{1}{p}$. We construct a QCCC KE $(\mathcal{A}, \mathcal{B})$ as follows:

1. $\mathcal{A}$ and $\mathcal{B}$ take 1^λ as input.
2. $\mathcal{A}$ runs $\mathcal{P}_1(1^\lambda)$, and $\mathcal{B}$ runs $\mathcal{V}_1(1^\lambda)$. Let $(\text{state}, \sigma_{\text{state}})$ be $\mathcal{P}_1$'s output. Let v be $\mathcal{V}_1$'s output.
3. $\mathcal{A}$ runs $\mathcal{P}_2(\text{state}, \sigma_{\text{state}})$, and $\mathcal{B}$ runs $\mathcal{V}_2(v)$, but $\mathcal{A}$ does not send y to $\mathcal{B}$.
4. $\mathcal{B}$ runs $y' \leftarrow \text{Ext}(v, \tau, x)$.
5. $\mathcal{A}$ outputs $a := y$, and $\mathcal{B}$ outputs $b := y'$.

Now we show that thus constructed $(\mathcal{A}, \mathcal{B})$ is a $(1 - \frac{1}{p}, 1 - \frac{1}{q})$ -QCCC KE. From Lemma 2.11, such a $(1 - \frac{1}{p}, 1 - \frac{1}{q})$ -QCCC KE can be amplified to obtain a QCCC KE.

$(1 - \frac{1}{p})$ -correctness is clear from Equation (64). Next, we show $(1 - \frac{1}{q})$ -security. For the sake of contradiction, we assume that $(\mathcal{A}, \mathcal{B})$ is not $(1 - \frac{1}{q})$ -secure. This means that there exists a non-uniform QPT adversary $\mathcal{E}$ such that

$$\Pr \left[\begin{array}{l} (v, (\text{state}, \sigma_{\text{state}}); \tau) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle \\ y \leftarrow \mathcal{E}(\tau, x) : \begin{array}{l} x \leftarrow \mathcal{V}_2(v) \\ y \leftarrow \mathcal{P}_2(\text{state}, \sigma_{\text{state}}, x) \end{array} \end{array} \right] > 1 - \frac{1}{q(\lambda)} \quad (65)$$

for infinitely many $\lambda \in \mathbb{N}$. From this $\mathcal{E}$, we can construct a pair $(\mathcal{P}_1^*, \mathcal{P}_2^*)$ of adversaries that breaks $(\alpha - \frac{1}{q}, 0)$ -soundness of the extractable PoQM as follows:

- $\mathcal{P}_1^*$: Run $(\text{state}, \sigma_{\text{state}}) \leftarrow \mathcal{P}_1(1^\lambda)$. Let τ be the transcript. Output τ .
- $\mathcal{P}_2^*$: Take τ and x as input, and run $e \leftarrow \mathcal{E}(\tau, x)$. Send e to $\mathcal{V}_2$.

$(\mathcal{P}_1^*, \mathcal{P}_2^*)$ breaks $(\alpha - \frac{1}{q}, 0)$ -soundness as follows:

$$\Pr[\top \leftarrow \langle \mathcal{V}_2(v), \mathcal{P}_2^*(\tau) \rangle : (v, \tau) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1^*(1^\lambda) \rangle] \quad (66)$$

$$= \Pr \left[\top \leftarrow \mathcal{V}_2(v, x, e) : \begin{array}{l} (v, (\text{state}, \sigma_{\text{state}}); \tau) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle \\ x \leftarrow \mathcal{V}_2(v) \\ e \leftarrow \mathcal{E}(\tau, x) \end{array} \right] \quad (67)$$

$$= \Pr \left[\top \leftarrow \mathcal{V}_2(v, x, e) : \begin{array}{l} (v, (\text{state}, \sigma_{\text{state}}); \tau) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle \\ x \leftarrow \mathcal{V}_2(v) \\ e \leftarrow \mathcal{E}(\tau, x) \\ y \leftarrow \mathcal{P}_2(\text{state}, \sigma_{\text{state}}, x) \end{array} \right] \quad (68)$$

$$\geq \Pr \left[\top \leftarrow \mathcal{V}_2(v, x, e) \wedge e = y : \begin{array}{l} (v, (\text{state}, \sigma_{\text{state}}); \tau) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle \\ x \leftarrow \mathcal{V}_2(v) \\ e \leftarrow \mathcal{E}(\tau, x) \\ y \leftarrow \mathcal{P}_2(\text{state}, \sigma_{\text{state}}, x) \end{array} \right] \quad (69)$$

$$= \Pr \left[\top \leftarrow \mathcal{V}_2(v, x, y) \wedge y \leftarrow \mathcal{E}(\tau, x) : \begin{array}{l} (v, (\text{state}, \sigma_{\text{state}}); \tau) \leftarrow \langle \mathcal{V}_1(1^\lambda), \mathcal{P}_1(1^\lambda) \rangle \\ x \leftarrow \mathcal{V}_2(v) \\ y \leftarrow \mathcal{P}_2(\text{state}, \sigma_{\text{state}}, x) \end{array} \right] \quad (70)$$

$$> \alpha(\lambda) - \frac{1}{q(\lambda)} \quad (71)$$

$$(72)$$

for infinitely many $\lambda \in \mathbb{N}$. Here, in Equation (71), we have used the union bound. This contradicts $(\alpha - \frac{1}{q}, 0)$ -soundness of the extractable PoQM. $\square$

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References

- [ACC⁺22] Per Austrin, Hao Chung, Kai-Min Chung, Shiuan Fu, Yao-Ting Lin, and Mohammad Mahmoody. On the impossibility of key agreements from quantum random oracles. In Yevgeniy Dodis and Thomas Shrimpton, editors, *CRYPTO 2022, Part II*, volume 13508 of *LNCS*, pages 165–194. Springer, Cham, August 2022. (Cited on page 4.)
- [AQY22] Prabhanjan Ananth, Luowen Qian, and Henry Yuen. Cryptography from pseudorandom quantum states. In Yevgeniy Dodis and Thomas Shrimpton, editors, *CRYPTO 2022, Part I*, volume 13507 of *LNCS*, pages 208–236. Springer, Cham, August 2022. (Cited on page 3.)
- [BCM⁺21] Zvika Brakerski, Paul Christiano, Urmila Mahadev, Umesh Vazirani, and Thomas Vidick. A cryptographic test of quantumness and certifiable randomness from a single quantum device. *Journal of the ACM*, 68(5):31:1–31:47, 2021. (Cited on page 1, 8.)
- [BGKM⁺23] Zvika Brakerski, Alexandru Gheorghiu, Gregory D. Kahanamoku-Meyer, Eitan Porat, and Thomas Vidick. Simple tests of quantumness also certify qubits. In Helena Handschuh and

- Anna Lysyanskaya, editors, *CRYPTO 2023, Part V*, volume 14085 of *LNCS*, pages 162–191. Springer, Cham, August 2023. (Cited on page 7.)
- [BLMP23] Marshall Ball, Yanyi Liu, Noam Mazon, and Rafael Pass. Kolmogorov comes to cryptomania: On interactive kolmogorov complexity and key-agreement. In *64th FOCS*, pages 458–483. IEEE Computer Society Press, November 2023. (Cited on page 11.)
- [BZ13] Dan Boneh and Mark Zhandry. Secure signatures and chosen ciphertext security in a quantum computing world. In Ran Canetti and Juan A. Garay, editors, *CRYPTO 2013, Part II*, volume 8043 of *LNCS*, pages 361–379. Springer, Berlin, Heidelberg, August 2013. (Cited on page 4, 6, 7.)
- [ÇG24] Alper Çakan and Vipul Goyal. Unbounded leakage-resilient encryption and signatures. Cryptology ePrint Archive, Paper 2024/1876, 2024. (Cited on page 4, 8.)
- [CH22] Nai-Hui Chia and Shih-Han Hung. Classical verification of quantum depth. *arXiv preprint arXiv:2205.04656*, 2022. (Cited on page 7.)
- [CR20] Rui Chao and Ben W Reichardt. Quantum dimension test using the uncertainty principle. *arXiv preprint arXiv:2002.12432*, 2020. (Cited on page 7.)
- [DFKP15] Stefan Dziembowski, Sebastian Faust, Vladimir Kolmogorov, and Krzysztof Pietrzak. Proofs of space. In Rosario Gennaro and Matthew J. B. Robshaw, editors, *CRYPTO 2015, Part II*, volume 9216 of *LNCS*, pages 585–605. Springer, Berlin, Heidelberg, August 2015. (Cited on page 1, 7.)
- [GMMY24] Eli Goldin, Tomoyuki Morimae, Saachi Mutreja, and Takashi Yamakawa. CountCrypt: Quantum cryptography between QCMA and PP. Cryptology ePrint Archive, Paper 2024/1707, 2024. (Cited on page 11.)
- [GV19] Alexandru Gheorghiu and Thomas Vidick. Computationally-secure and composable remote state preparation. In David Zuckerman, editor, *60th FOCS*, pages 1024–1033. IEEE Computer Society Press, November 2019. (Cited on page 5, 10.)
- [KT24] Dakshita Khurana and Kabir Tomer. Commitments from quantum one-wayness. In Bojan Mohar, Igor Shinkar, and Ryan O’Donnell, editors, *56th ACM STOC*, pages 968–978. ACM Press, June 2024. (Cited on page 1, 3.)
- [KT25] Dakshita Khurana and Kabir Tomer. Founding quantum cryptography on quantum advantage, or, towards cryptography from #p hardness. In Michal Koucký and Nikhil Bansal, editors, *57th ACM STOC*, pages 178–188. ACM Press, June 2025. (Cited on page 6, 8, 9.)
- [LLLL24] Longcheng Li, Qian Li, Xingjian Li, and Qipeng Liu. How (not) to build quantum PKE in minicrypt. In Leonid Reyzin and Douglas Stebila, editors, *CRYPTO 2024, Part VII*, volume 14926 of *LNCS*, pages 152–183. Springer, Cham, August 2024. (Cited on page 4.)
- [LLLL25] Longcheng Li, Qian Li, Xingjian Li, and Qipeng Liu. Toward the Impossibility of Perfect Complete Quantum PKE from OWFs. In Raghu Meka, editor, *16th Innovations in Theoretical Computer Science Conference (ITCS 2025)*, volume 325 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 71:1–71:16, Dagstuhl, Germany, 2025. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. (Cited on page 4.)

- [LLQ22] Jiahui Liu, Qipeng Liu, and Luowen Qian. Beating classical impossibility of position verification. In Mark Braverman, editor, *13th Innovations in Theoretical Computer Science Conference, ITCS 2022, January 31 - February 3, 2022, Berkeley, CA, USA*, volume 215 of *LIPICs*, pages 100:1–100:11. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2022. (Cited on page 5, 6, 9, 10.)
- [MM25] Giulio Malavolta and Tamer Mour. How to verify that a small device is quantum, unconditionally. Cryptology ePrint Archive, Paper 2025/970, 2025. (Cited on page 7.)
- [MSY25] Tomoyuki Morimae, Yuki Shirakawa, and Takashi Yamakawa. Cryptographic characterization of quantum advantage. In Michal Koucký and Nikhil Bansal, editors, *57th ACM STOC*, pages 1863–1874. ACM Press, June 2025. (Cited on page 3.)
- [MV20] Jack Maxfield and Thomas Vidick. Quantum proofs of space, 2020. (Cited on page 1, 4, 7.)
- [MY22] Tomoyuki Morimae and Takashi Yamakawa. Quantum commitments and signatures without one-way functions. In Yevgeniy Dodis and Thomas Shrimpton, editors, *CRYPTO 2022, Part I*, volume 13507 of *LNCS*, pages 269–295. Springer, Cham, August 2022. (Cited on page 3.)
- [RS19] Roy Radian and Or Sattath. Semi-quantum money. In *Proceedings of the 1st ACM Conference on Advances in Financial Technologies, AFT 2019, Zurich, Switzerland, October 21-23, 2019*, pages 132–146. ACM, 2019. (Cited on page 5.)
- [Yan22] Jun Yan. General properties of quantum bit commitments (extended abstract). In Shweta Agrawal and Dongdai Lin, editors, *ASIACRYPT 2022, Part IV*, volume 13794 of *LNCS*, pages 628–657. Springer, Cham, December 2022. (Cited on page 3.)
- [Zha25] Jiayu Zhang. Formulations and constructions of remote state preparation with verifiability, with applications. In Raghu Meka, editor, *16th Innovations in Theoretical Computer Science Conference, ITCS 2025, January 7-10, 2025, Columbia University, New York, NY, USA*, volume 325 of *LIPICs*, pages 96:1–96:19. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025. (Cited on page 5, 10, 11.)